



European University Alliance for Global Health

# eduTAP

<https://eduTAP.eu>

*A Concept and System of Digital Identity Credentials for proximity on-site service access*

Modern Campus ID (Student ID  Staff ID) for universities in Europe & the world

Alexander Loechel – PloneConf 2025

PyCon Finland – University of Jyväskylä Finland



Co-funded by the Erasmus+ Programme of the European Union

[www.eugloh.eu](http://www.eugloh.eu)

# Digital Credentials for Higher Education

Two Digital Credential Verticals, one Wallet Environment

## Long-Term Digital Credentials



## European Digital Identity Wallet (EUDI-Wallet)



EU Digital Identity  
**Wallet**



## Short-Term Digital Credential



- Diplomas / Diploma Supplements
- Transcript of Records
- Micro-Credentials

- Container for Credentials
- Web-Wallet

- Student IDs
- Library-IDs
- Event-Tickets

→ Digitization of Academic Proofs of Achievement

→ Digitize, **digitalize** and simplify administrative processes

→ Digitalization of Service Access

→ DC4EU / EBSI / EDEH Interoperability Framework

→ European Student Card / **eduTAP**

# Stages of Digitalization – Period of Digital transformation

## Digitalization 1.0

- Digitization – create digital versions of analog documents

→ Digitization vs. Digitalization (machine readable content / payload)

## Digitalization 2.0

- Digitalization with internal focus
- Optimize internal administrative processes – **organization centric optimizations**
- Processing of data

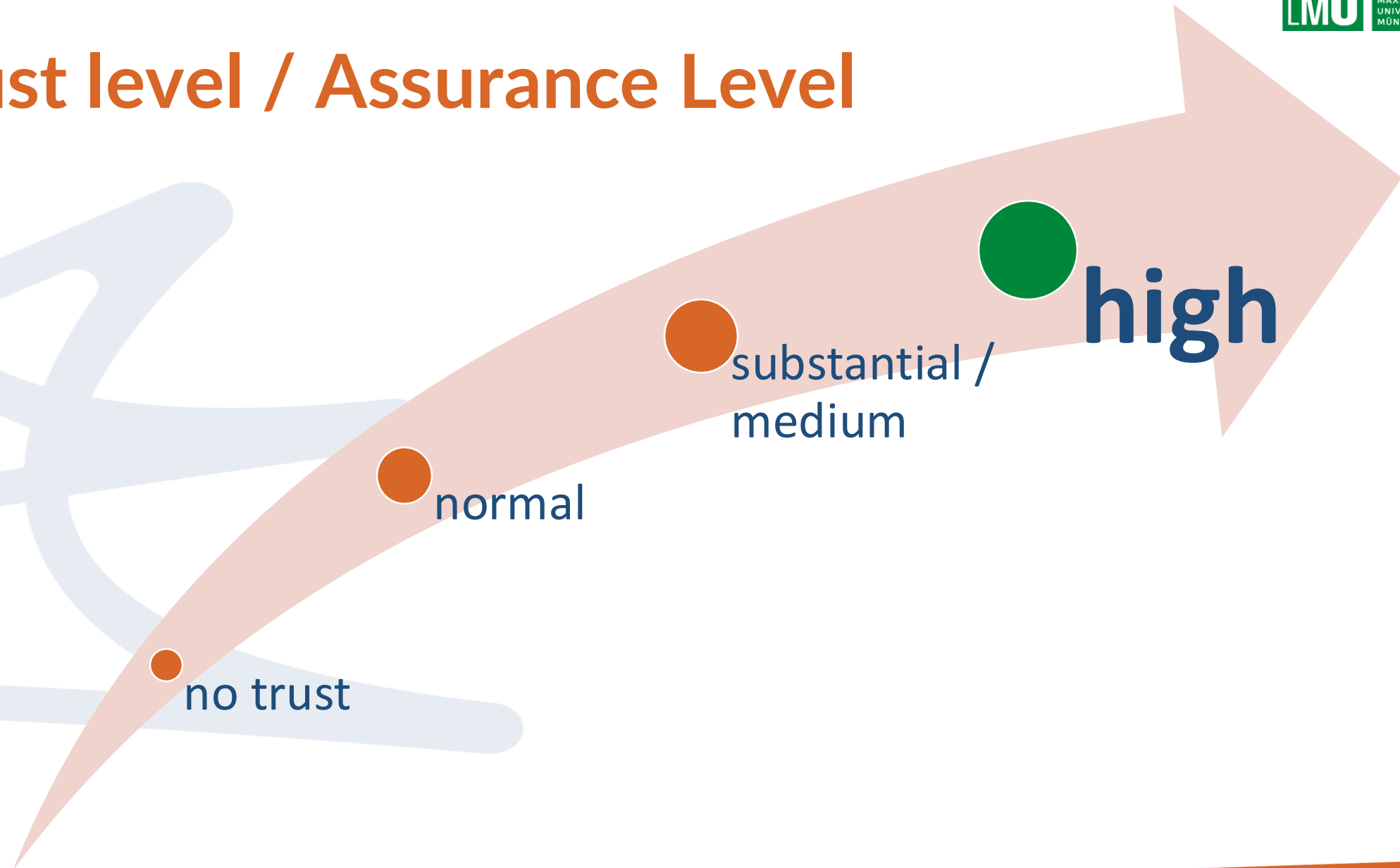
→ Digital transformation (a hard shift)

## Digitalization 3.0

- Digitalization with focus on the user – **user centric optimizations**
- Optimization on how services are discovered and seamlessly used
- Flow of data

→ Service-oriented  
→ **Digital identity**  
→ Interoperability

# Trust level / Assurance Level



# Recognition of eduTAP @ Conferences

## ECCA Conference 2025 LMU Munich, Germany

**Mikhail Strunkin**  • 1.  
Global Physical Access Partnerships for Google Wallet at Google  
1 Monat • Bearbeitet • 

That time of year again - when tech vendors and universities gather under the roof of **ECCA - European Campus Card Association** to tackle the never-ending puzzle of digitizing the European Student Card.

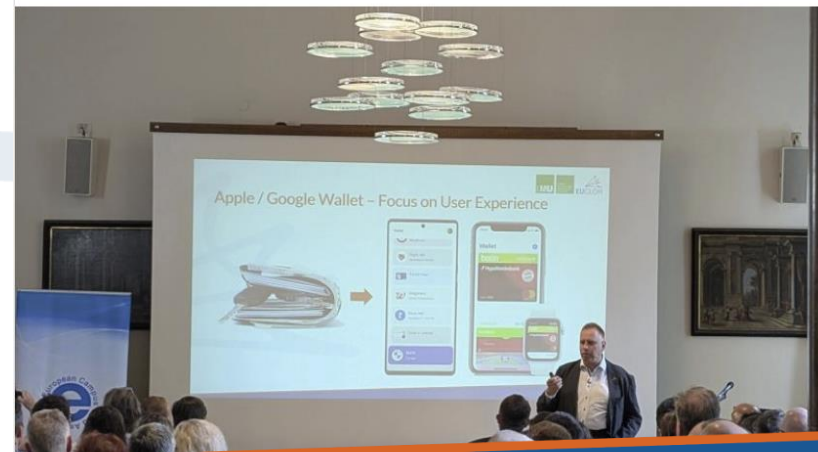
This year, it feels like **Alexander Loechel** might just be the one to finally crack the code!

Let's see if 2025 is finally the year we go fully digital at **Ludwig-Maximilians-Universität München** and other **EUGLOH** members.

I have some confidence, finally :)

**#DigitalStudentID #ECCA2025 #eduTAP #GoogleWallet**

Übersetzung anzeigen



## EUNIS Conference 2025 Belfast, UK → Elite Award

**EUNIS**  
846 Follower:innen  
3 Wochen • 

🎉 Celebrating Excellence at #EUNIS25! 🎉

We are proud to announce the recipients of this year's EUNIS Awards, recognizing outstanding contributions to digital transformation in higher education across Europe:

- 🏆 Best Paper Award went to **Bernd Decker** and **Denise Dittrich** (IT Center, RWTH Aachen University)  
📄 Paper: "Cost Management for the Use of Generative AI in Higher Education: A Case Study from North Rhine-Westphalia"
- 🏆 Elite Award awarded to **Simon Lund** and **Alexander Loechel** (Ludwig-Maximilians-Universität München)  
📄 Paper: "eduTAP Common ID - A Standardized Digital Identity Document for HEIs"
- 🏆 Special Award on lasting achievement was granted to **Ognjen Orel**, **Mattija Kranjčič**, and **Mirna Imrović** (University Computing Centre, University of Zagreb):  
📄 Paper: "The Evolution of ISVU: 20+ Years of the National Student Management System in Croatia"
- 🏆 Darup Award recognised the **le Conservatoire National des Arts et Métiers** team for their innovative work  
📄 Paper: "When Virtual Reality Reinvests Training: The Cnam Bet with CAP'VR"
- 🏆 Congratulations to **Christian Cousquer**, **Maïté SYLLA**, **Rebeca Garcia**, **Sohayb Khoulani**, **Fanny Haugulier**, **Catherine Gomez**, **Vincent Caqueret**, **Marion Pommet**, and the entire team!
- 🏆 Emerging Professionals Award  
Awarded to **HANBIT CHOI**, **Aksel Skullerud Fosaas**, **Zara Mudassar**, **Sander Garland Slinning**, **Arman Bulak Hagella**, and **Eira Bjerkeng Scherjon** (**Sikt - Kunnskapssektorens tjenesteleverandør**)  
📄 Paper: "Beyond the Hype: Real-World Insights on AI Integration and the Value of Alternative Solutions"

🎉 A huge congratulations to all winners for your innovation, dedication, and impact on the higher education IT landscape!

**#EUNISAwards #HigherEd #DigitalInnovation #AI #VR #DigitalID #EUNISCom**

Übersetzung anzeigen





European University Alliance for Global Health

# eduTAP

<https://eduTAP.eu>

*Modern Campus ID (Student ID / Staff ID)  
for universities in Europe & the world*

*Concept of eduTAP*





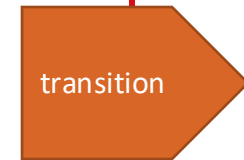
# The LMU Student ID (over time)

LMU Student ID till 2019

LMUcard Student ID 2019-2025

LMU Student ID 2025+

LMU Student ID 2027+ ?



today

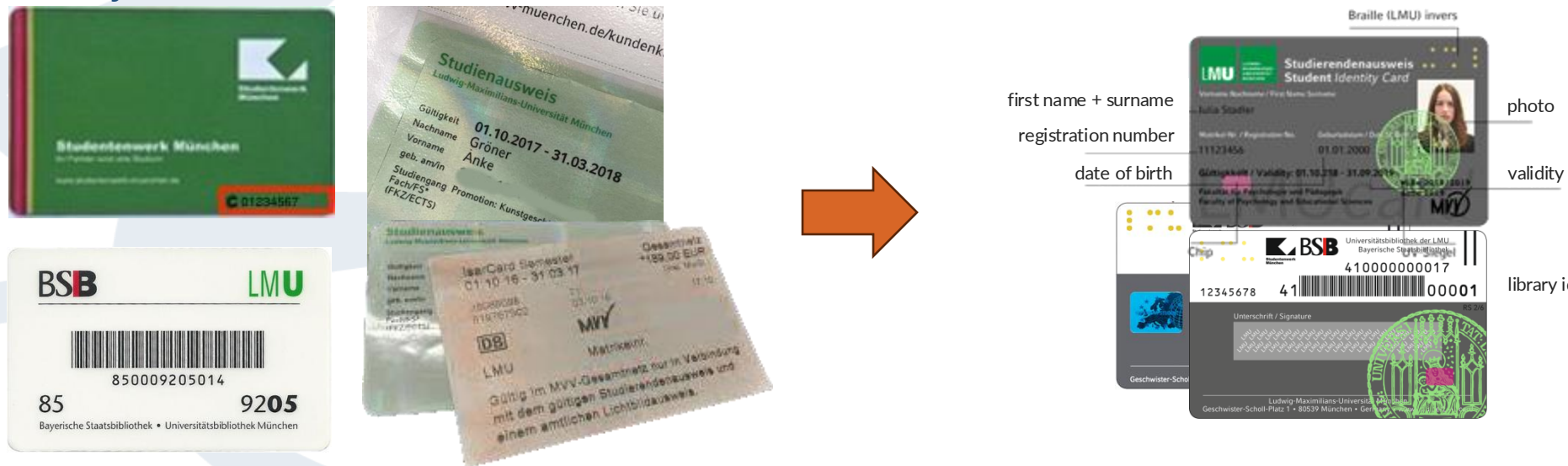


eduTAP & eduTAP@LMU

# LMUcard – LMU Student ID

The transition of LMU Student IDs from paper to plastic cards (LEGIC CTC4096-MM410)

→ A Project 2015 – 2019



**Goal:** A modern integrated, contactless card for all services at LMU ← Better User Experience

- ✓ Visual Student ID – contactless readable registration number
- ✓ Payment at mensa / canteens, cafeterias and vending-machines, copiers
- ✓ Library ID ✓ Public Transport Ticket (MVV Semesterticket)

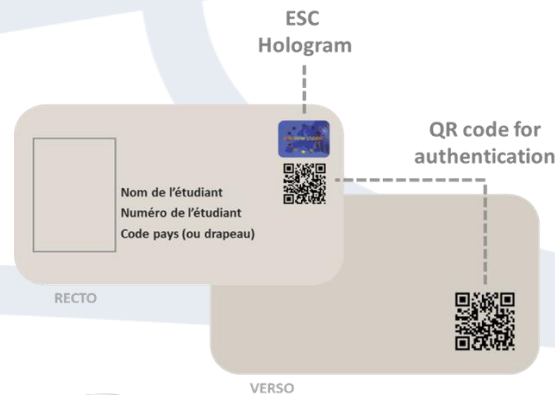
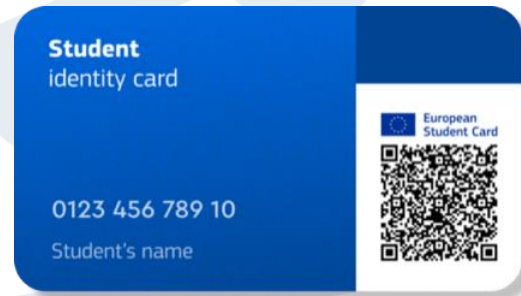
*It was a hard project with a lot of stakeholders*



# The European Student Card (ESC)

The **European Student Card** – a project to add a **European dimension** to local Student IDs

✓ Securely validate the student status



Idea by →  
Jean-Paul Roumegas  
(les Crouse)

## European Student Card logo

*Combines the European identity (text & flag) with the QR code. It facilitates the implementation and recognition of the ESCs*

## ESCN - European Student Card Number

*Unique card identifier that serves to link a specific card to the specific higher education institution who produced it and to identify a card*

## ESI - European Student Identifier

*Unique identifier that allows to link a given student to a specific ESC. It is not recommended to be printed on the card.*



**European Student Card**



✓ A great vision



# EUGLOH - The European University Alliance for Global Health

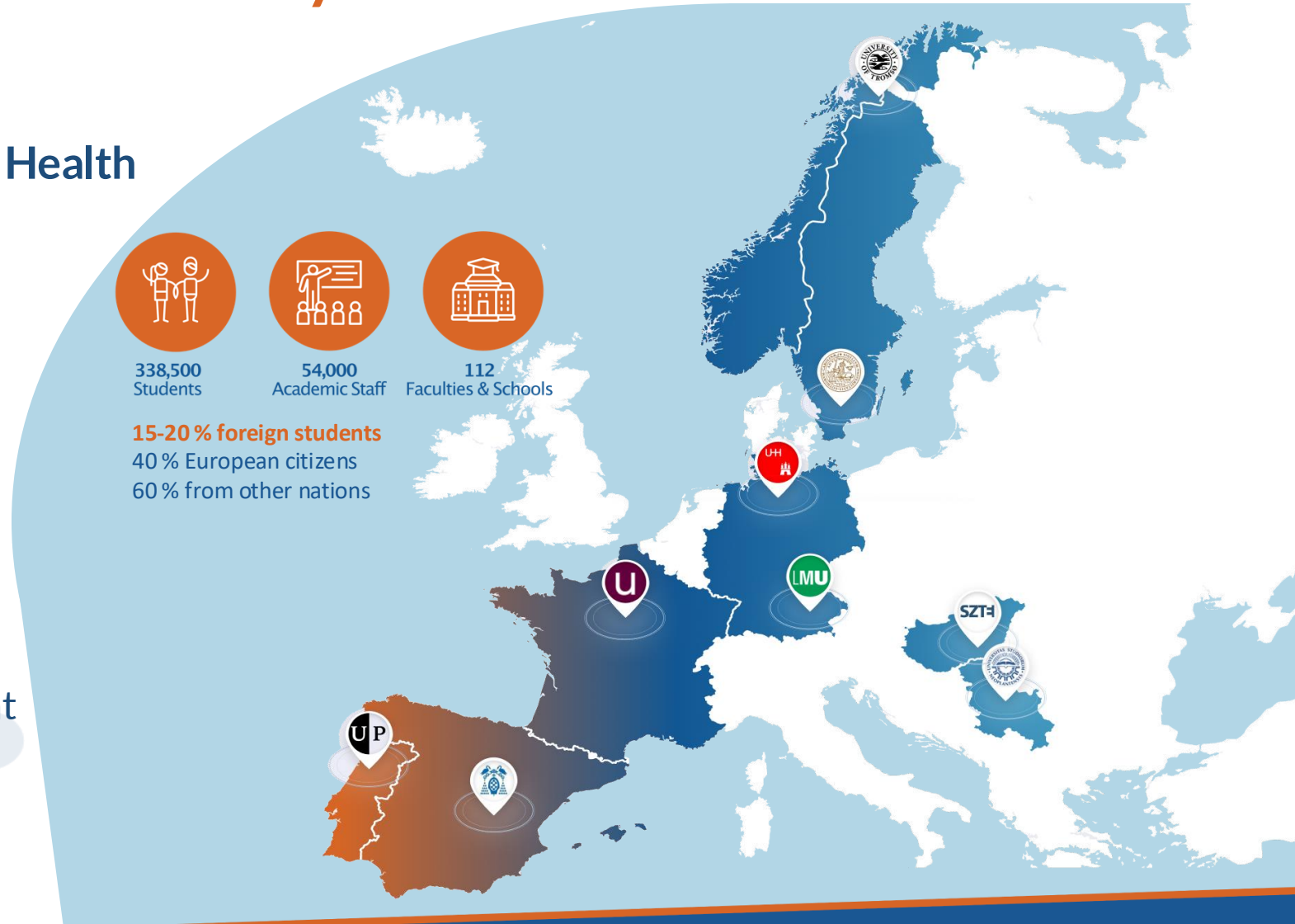
## 9 Universities, 1 Goal

### Building Europe's Campus for Global Health

- Equipping students and staff with the skills needed to shape the future of global health
- Creating a **seamless, inclusive European campus** that enables **mobility**, joint learning opportunities, and **shared infrastructure**
- Addressing pressing health issues through interdisciplinary, co-created education, research and innovation that engages with social needs



15-20 % foreign students  
40 % European citizens  
60 % from other nations



# LMUcard + European Student Card (ESC)

## LMU as an international university

## International Partners and the European University Alliance

### Strategic partnerships

- LMU-Bordeaux Research Cooperation Program
- Cambridge LMU Strategic Partnership
- LMU-China Academic Network (LMU-ChAN)
- LMU-NYU Research Cooperation Program
- LMU-TAU Research Cooperation Program
- LMU-Todai Cooperation in the Sciences
- LMU-UCB Research in the Humanities

### EU projects – The European University Alliances

European University Alliance for Global Health (EUGLOH)



### Global collaborations

- LMU - China Scholarship Council (CSC) Program
- LMU-Harvard Young Scientists' Forum
- LMU's Latin America Network



### International alliances

- League of European Research Universities (LERU)
- European University Association (EUA)
- Europaeum



*Infinity partners → infinity stakeholders → infinity systems*

# A shared vision – seamless mobility

EUGLOH & the European Student Card have a **shared vision**

- ✓ Seamless mobility
- ✓ Easy service access

For:

- ✓ students,
- ✓ staff and
- ✓ professionals



EUGLOH ESC Working Group  
formed end of 2020:

**All EUGLOH universities should adopt ESC**  
and define services to share (Lead Paris-Saclay).

But differ in the audience and baseline  
technical requirements

# The European Student Card (2020)

## Basic concepts

The European Student Card (ESC) is a new standard created by European Higher Education Institution (HEI) to promote student mobility within Europe (EHEA zone).

→ The long-term objective is to give all European Students the same rights as a local student when travelling abroad.

So, the idea is to create a card that can be accepted anywhere within Europe. This card gives access to an unlimited number of services that students use in their daily life, such as borrowing book from university library, accessing student cafeteria with discounted prices, accessing university premises, using public transportation etc.



## Short- and long-term perspectives

The first basic service to be set up is **student status authentication**. This service offers the possibility for any organization (private business, public administration, association, HEI...) to confirm whether a card holder is a student or not. **This service is a necessity because student status authentication is usually a prerequisite to give access to other services that are restricted to students.**

- Other services will **become available afterward**, such as:
  - The possibility to use the card as an **electronic wallet** for day-to-day expenses
  - The possibility to write new information on the card after it has been delivered, granting access to a wide range of new services for the student
  - The possibility of using the card as a **key** to allow **exchange of data between HEIs**, to facilitate administrative procedures during student mobility
- **Dematerialisation of the card itself, using a smartphone**

Source: ESC Handbook for institution: Users's Guide to connect to the Platform  
(<http://europeanstudentcard.eu/wp-content/uploads/2017/02/ESC-User-Guide-def.pdf>)



# EUGLOH campus cards state and connected services

## → Vision:

All university members should be able to use services of host university while on visit → easy and contactless



## Status of four core members (IT wise - 2020)

### Ludwig-Maximilians-Universität München

- Legic CTC Cards (Legic Advant + Prime)
- Payment (Studierendenwerk München)
- Cloud-Printing and Scanning (Secure-Printing and Payment)
- Library
- Ticket for Public Transport System

### Université Paris-Saclay

- NXP Mifare DESfire Cards
- Payment (izly – les Crous)
- Library
- Physical Access Control

### Lunds universitet

- NXP Mifare Classic Cards
- Physical Access Control (online and offline)
- Login at copiers (for copy/print/scan)
- Library
- Several other services, but not any payment services

### Universidade do Porto

- NXP Mifare Classic Cards / Java Cards with Mifare Classic emulation
- Library
- Physical Access Control
- Canteen & cafeterias
- Printing Service
- Attendance

All contactless, but  
Not compatible technologies



# EUGLOH & ESC – a shared vision and ambitions

We have a shared vision: **seamless mobility & easy service access**

But an ESC QR-Code was no option for us (GDPR & IT-Security wise), the long-term perspective a digital smartphone pass looked fascinating



→ a great vision



EUGLOH **wanted to share their services** across the alliance, and beyond (as we all have more partners)



The IT-Persons from LMU, U.Porto and Lund already knew each other → we met at ECCA 2019

- ✓ **understand the technological possibilities,**
- ✓ **knew and understand the topics, have access to major technology providers**
- ✓ **but also, the capabilities of the ESC Project team**

→ *If we want a good and interoperable solution in the near future, we must come up with a working proposal*

→ *That meets our requirements and audience (students, staff and professionals)*

# Campus Cards are about Accessing Services

## Identification

- **Identification**
- **Status verification**
- **Proof of entitlement**
- Single Sign On / 2FA / FIDO2
- Attendance check
  - Check-in for exams
  - attendance at courses
  - Time recording
- Electronic Signature (of legal documents)

## Electronic payment / cashless campus

- Canteen & cafeterias
- Vending machines
- Printing / scanning (secure & follow me printing)
- Ticketing (events and conferences)

## Physical Access Control

- Areas (campus, parking lot)
- Buildings
- Rooms
  - Classrooms
  - Labs
  - Computer rooms
  - Learning spaces
  - Offices
  - Accommodation facilities (i.e., dorms)
- Athletic facilities

## Library services

- Access to / borrow
  - Physical media (book, audio and video media)
  - E-media (book, audio and video media)
- Learning spaces

## Transport

- On campus services (university shuttle service)
- Public transport tickets & discounts

## Discount and promotions

- **Discounts** on cultural activities
  - Museums
  - theaters
  - cinemas
- Shops
- Restaurants

On- & Off-Campus → On-Site Usage / Proximity Use-Cases  
+ Online Services



# Campus Cards from a User Experience Point of View

Campus Cards (Student IDs / Staff IDs / Affiliate IDs) function as:

- **Official (visual) Identity Documents** and for university branding
- Primary medium for **Service Access** on-site → **proximity use-cases**
- Service Domains:
  - On-Campus
    - **Identification** and attendance tracking (courses, internships, exams)
    - **Payment** (open loop or closed loop payment) and discounts eligibility
    - **Physical Access Control** (buildings, labs, dorms)
    - **Library Services**
    - on-campus & public **Transport** integration (regional semester ticket models)
  - Off-Campus (Museums, theaters, shops, restaurants, ...)
    - **Status-based Identification** for third-party services
    - **Eligibility Verification** (e.g., student status, age-based discounts)



# Campus Cards are about Accessing Services

## The Service Domains

- Payment / Cashless Campus,
- Physical Access Control ,
- Library services,
- Transport

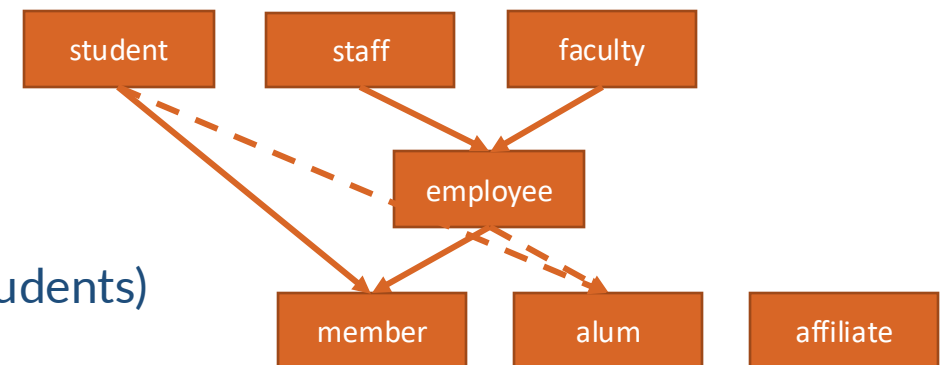
are “**local**” services, which only needs to be solved on a local level

- ✓ Separate service passes can choose best fitting technology
- ✓ Services are provided to **ALL** persons (not only one role, e.g., students)

→ Student IDs / Staff IDs are **legally limited** to enrolled / employed members of a Higher Education Institution (HEI)

→ Dedicated service passes can be legally issued to not enrolled / employed persons ← Users of the service

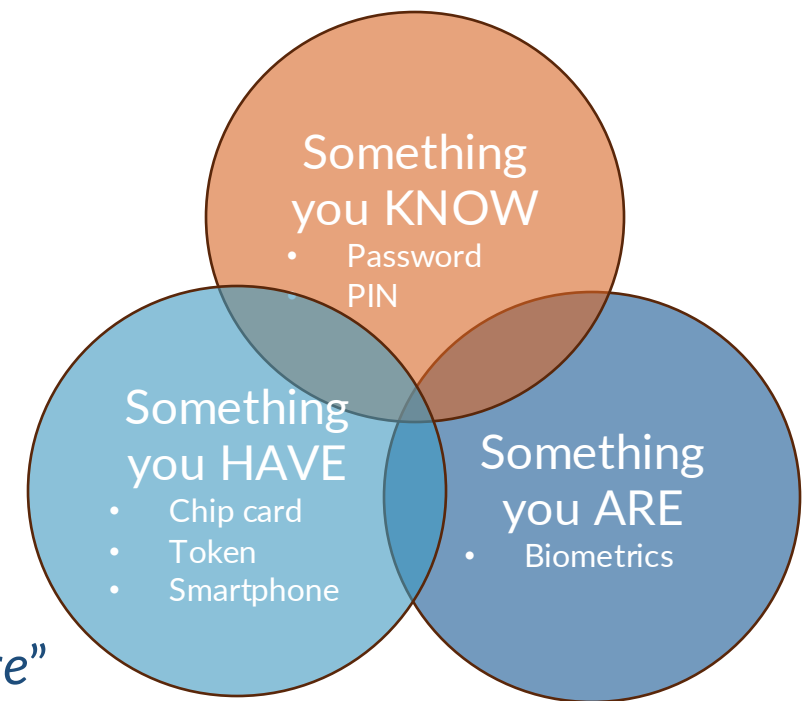
## Educational role system (from eduPerson)



# Key to all Services → Identification / Authentication

- Triple-A-System
  - (Identification)
  - **Authentication**
  - **Authorization**
  - **Accounting**
- **On-site services access** is about proving your entitlements and identity
- “*something you know*” as an authentication method for on-site service access is not very **efficient**, best a combination of “*something we have*” + “*something we are*” → Smartphone Wallet passes + authentication via biometrics
- We already have a powerful IDM System for **online-services** → Shibboleth → eduGAIN-AAI

## Authentication Factors





# Trust level / Assurance Level

- Very important for **IDENTIFICATION & AUTHENTICATION**
- Quality of identity (personal) data
- Type of authentication & consequences for identification medium

no trust

normal

Authentication:  
 • username / email  
 + password  
 Identity proofing:  
 → self-asserted  
 → verified e-mail address

substantial / medium

Authentication  
 • certificate based  
 • FIDO2  
 • Smart ID  
 Identity proofing:  
 → officially confirmed  
 → manual check of a government issued photo-ID

high

Authentication  
 • **eID.AS** compliant  
 → eID of EU Citizen IDs  
 → provided by HEI for foreign Students  
 Identity proofing:  
 → officially confirmed  
 → Identity document automated and technically checked to be genuine and represent the claimed identity

# DRY: do not re-invent the wheel

A complex system that works is invariably found to have evolved from a simple system that worked. The inverse proposition also appears to be true:

A **complex system designed from scratch never works and cannot be made to work**. You have to start over, beginning with a working simple system.

Source: "John Gall's law" - from "Systemantics: How Systems Really Work and How They Fail" - 1975

→ Digitalization Projects are complex enough, don't try to build everything from scratch

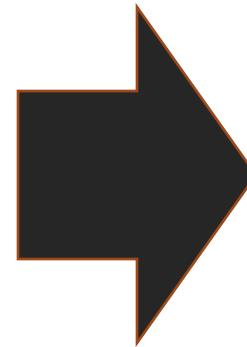
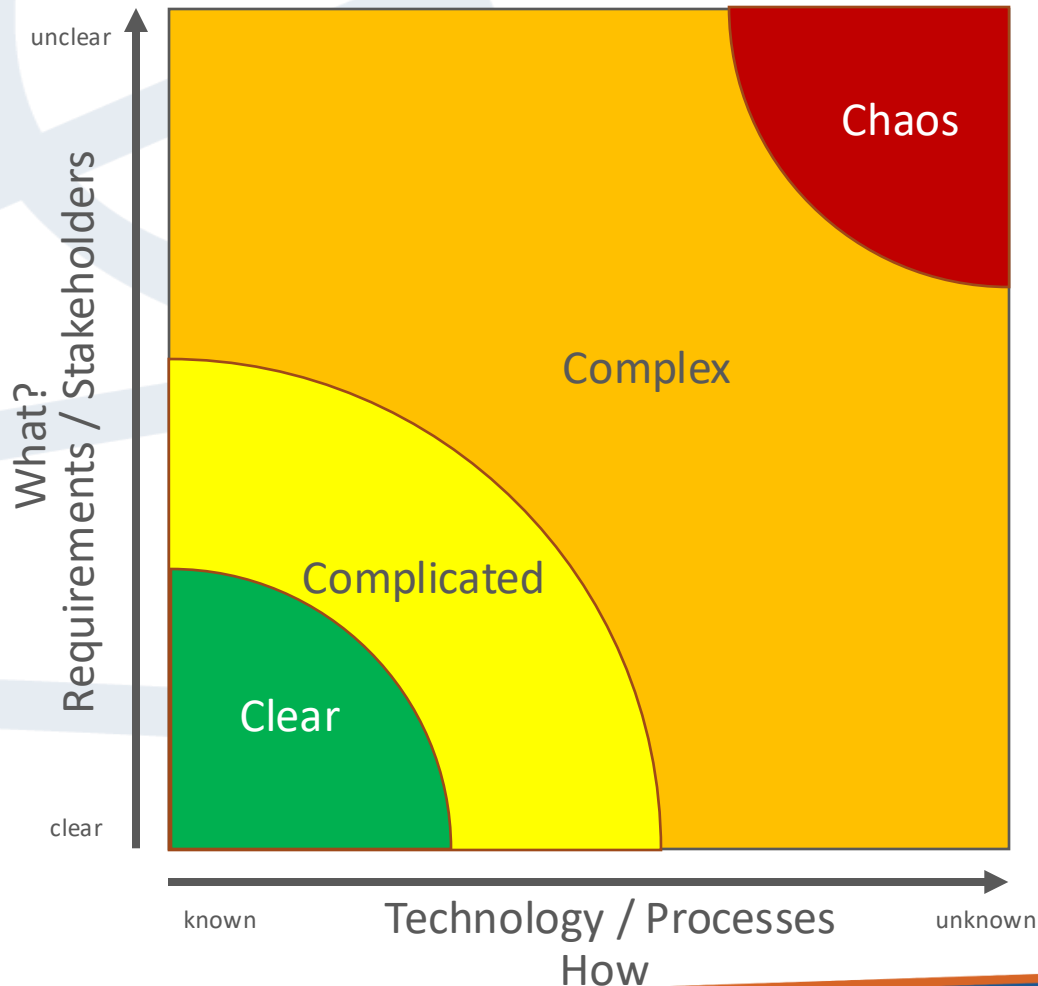
*If I have seen further it is by standing on the shoulders of Giants.*

Source: "Sir Isaac Newton letter to Robert Hooke dated 5 February 1675"



# Success factors – Understand the complexity

→ Complexity management and possibilities to act – Stacey Matrix & Cynefin-Framework



How to KISS: Keep it Simple, Stupid





# Munich campus as an Example of SHARED Services



Most European Universities do not have a clearly demarcated campus area. They are just in the city.

The university is not even the main service provider for student life.

Typical Service-Providers:

-  (=Student Affair Union)
  - Canteen & cafeterias
  - Vending machines
  - Student housing / dorms

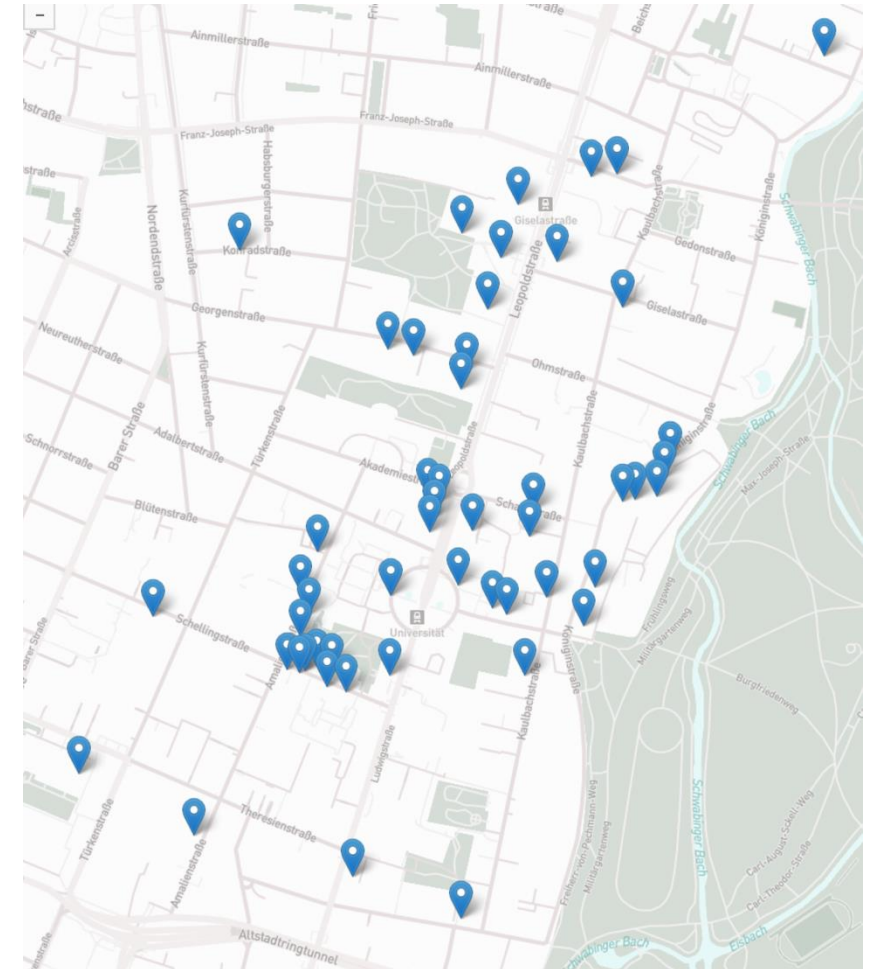
→ Responsible for all universities in Munich (LMU, TUM, HM, ...)
- Libraries are dedicated (almost independent) units at LMU we have two libraries:



**BSB** Bayerische Staatsbibliothek

Information in erster Linie

Bavarian State Library
- Transport → Public Transport System





# Campus Cards are about Accessing Services

Almost all services are **SHARED** services

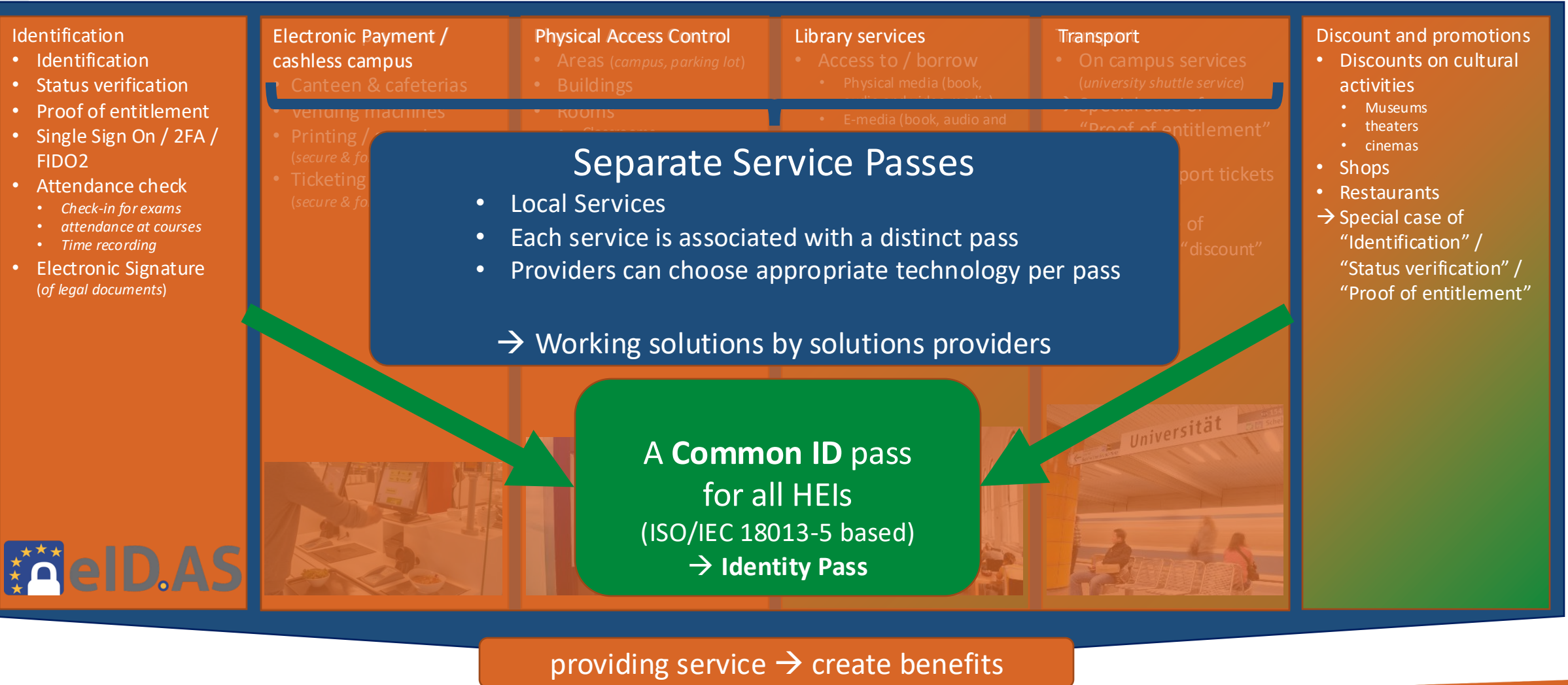
- ✓ University library ← Can be used by members of all universities in town / metropol area, in a university alliance or are open to the public
- ✓ Transport (semester ticket) ← Are just discounted and time limited public transport tickets (valid for 6 month)
- ✓ Canteens, cafés and vending machines on campus ← Can be used by all roles and members of all universities
- ✓ Physical Access Control (PAC) ← For all roles in a university / university alliance

Services are provided by **different Service Providers** (European universities are not business focused)

- ✓ Canteens, cafes and vending machines are operated by Student Affairs Unions
- ✓ Transport services are operated by local transport companies
- ✓ University Buildings (PAC) are university or state owned

→ Each **Service Provider** can issue their **own pass**, matching their needs and security requirements

# Think outside the card: Medium → Services



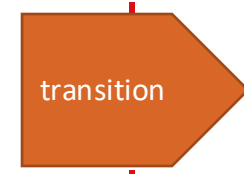
# The LMU Student ID (over time)

LMU Student ID till 2019

LMUcard Student ID 2019-2025

LMU Student ID 2025+

LMU Student ID 2027+ ?



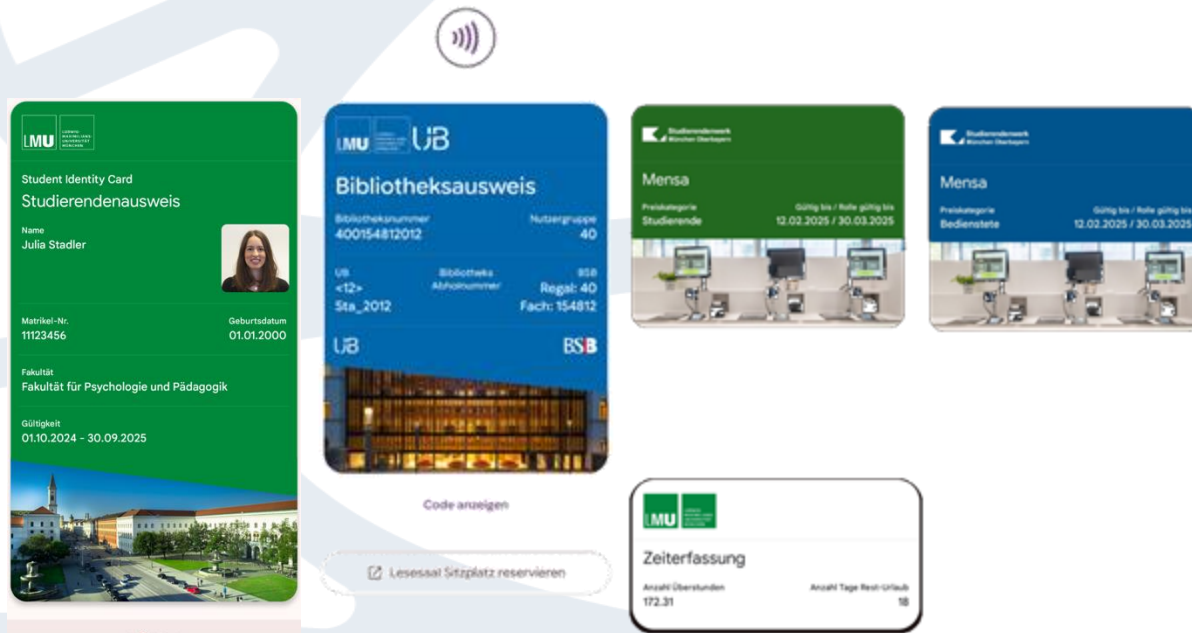
today

eduTAP & eduTAP@LMU

# Transition of credentials into the Smartphone Wallets

## Examples: Active and potential IDs for LMU Munich

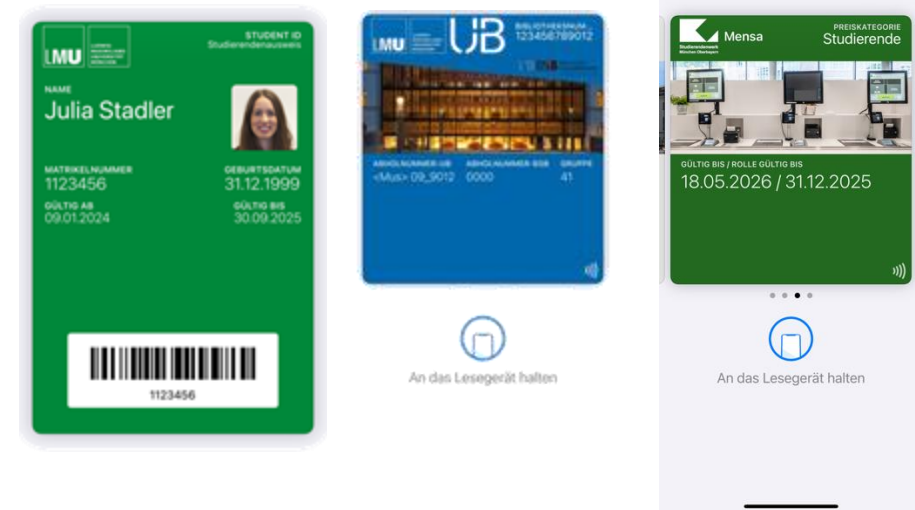
### Google Wallet



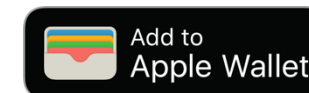
→ all **NFC enabled**



### Apple Wallet

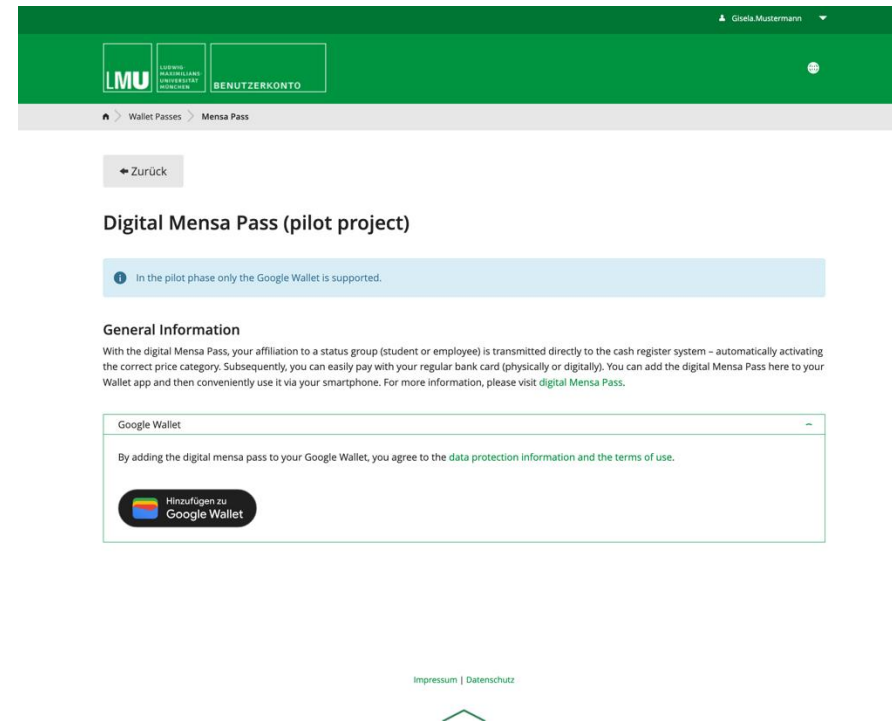
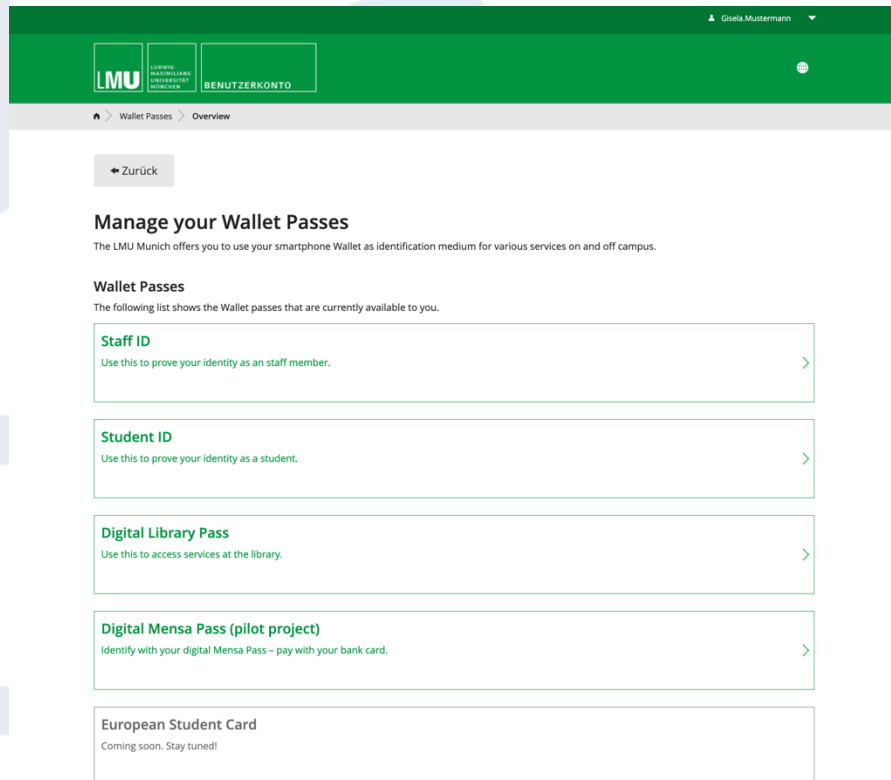


→ **NFC enabled** where possible

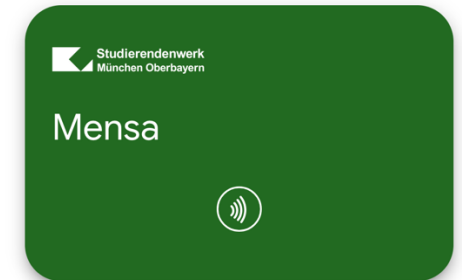


# Transition of credentials into the Smartphone Wallets

## Examples: LMU Munich Benutzerkonto – Pass Portal



Add loyalty card



After adding your loyalty card to Wallet, you'll see your point balance and loyalty benefits in places like Maps, Shopping, and more. You can turn this off in [Google Wallet passes data](#) or on an individual pass in the Google Wallet app.

Pass providers can automatically add related passes, promotions, offers and more to your existing passes which can be controlled on your device in Wallet settings.

[Terms and Privacy Policy](#)

Add

→ Plan to have all pass types online at LMU in the summer semester 2026 mostly design questions and information on the back are discussed



# Transition of credentials into the Smartphone Wallets

## LMU Student ID

The new **LMU Student ID** in the Wallet

- ✓ will come operational in summer semester 2025 (optional to LMUcard)
- ✓ for summer semester 2027 either physical LMUcard or wallet ID



Google



Apple

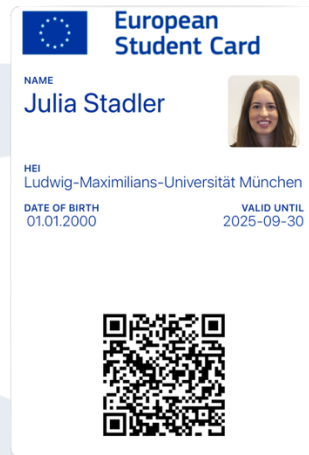
more wallets to come

# Transition of credentials into the Smartphone Wallets The European Student Card

## The European Student Card (v1.1 new ESC Design)

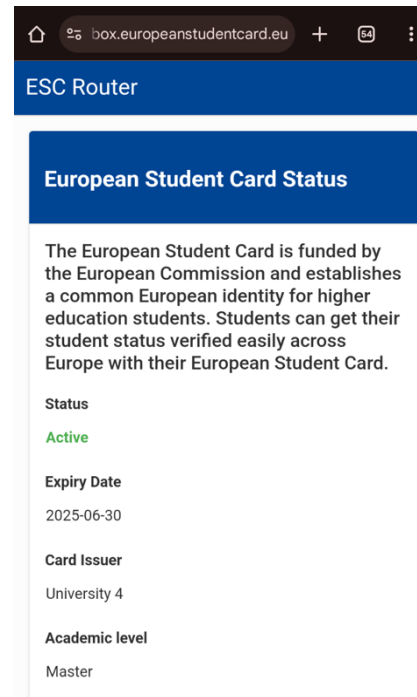


Google



Apple

- ✓ Discounts off-campus
- ✓ European dimension → Erasmus+ marketing



(v1.5)



QR-Code

# Transition of credentials into the Smartphone Wallets

## Library ID

NFC enabled by default



Code anzeigen

Google

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An das Lesegerät halten

Apple

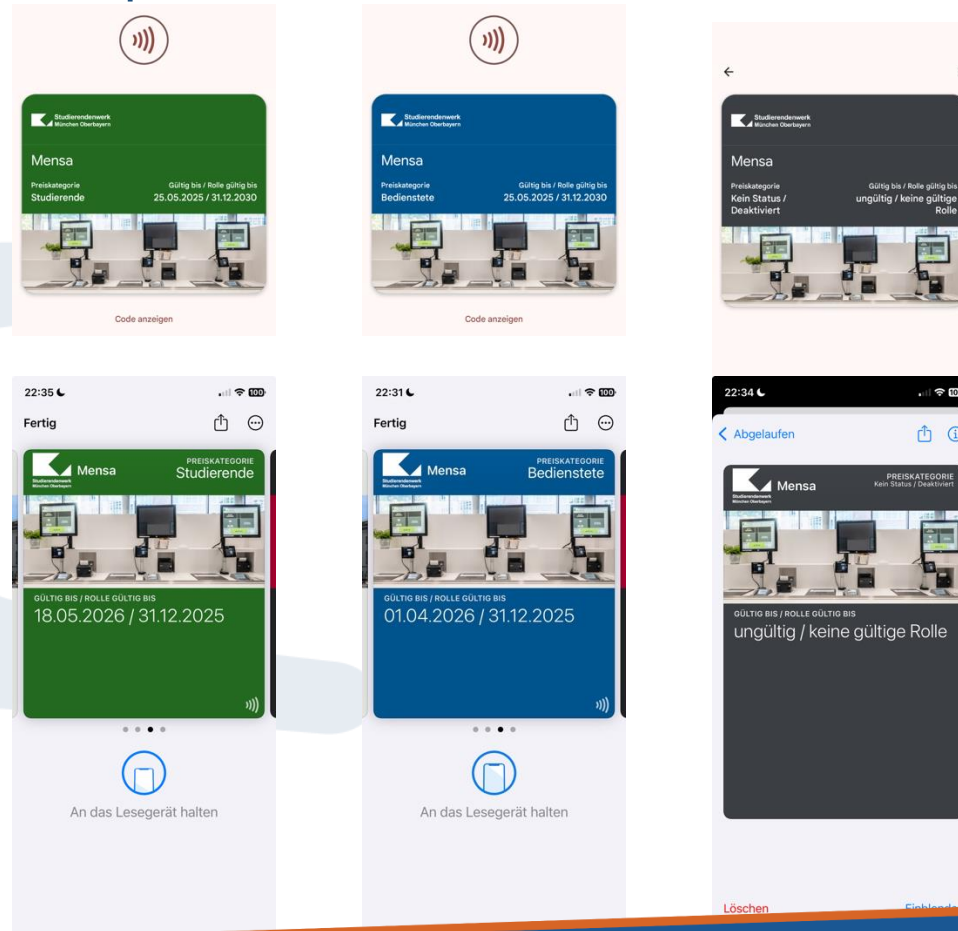


# Discount-Pass for Mensa Studierendenwerk München Oberbayern



Eligibility verification for discounts with the Apple / Google Wallet via NFC, payment with an Open-Loop Bank / Credit-Card

Google Wallet  
Since Feb. 2025



Apple Wallet



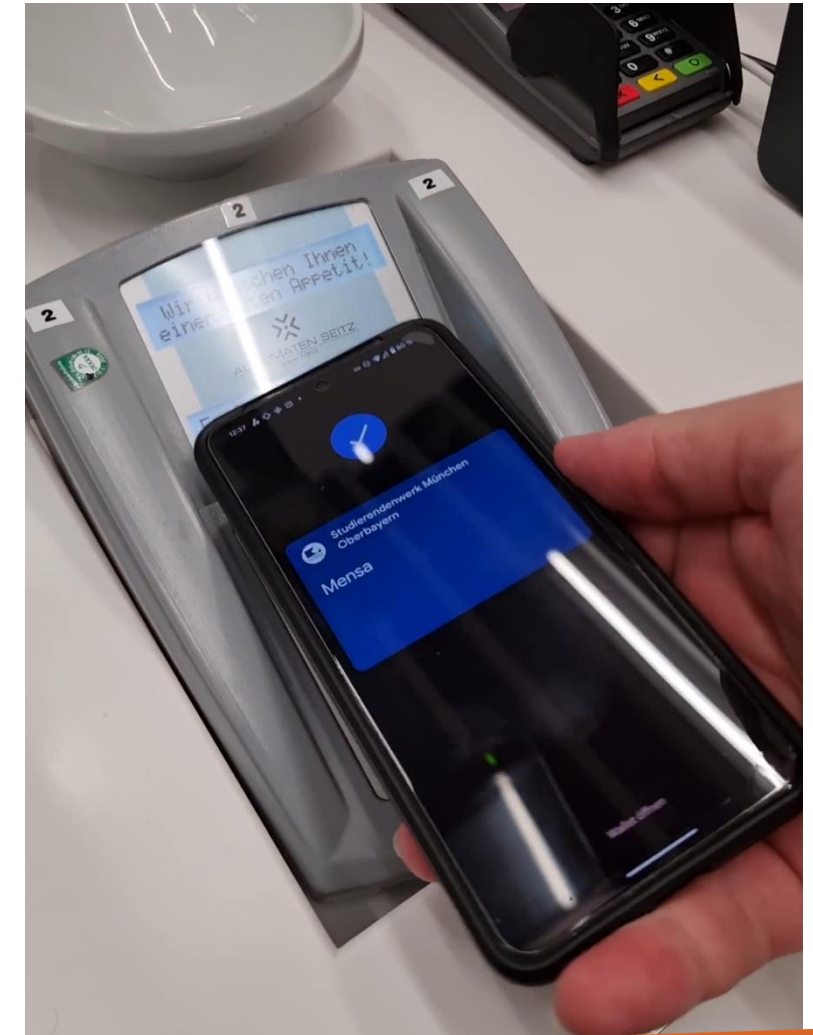
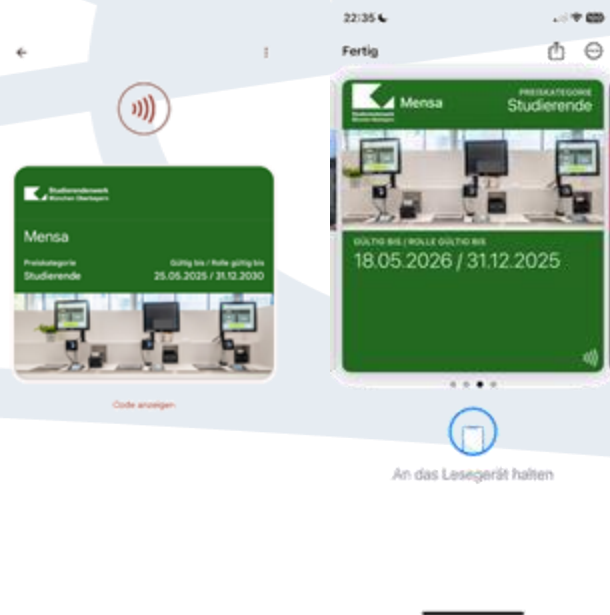
# Discount-Pass for Mensa Studierendenwerk München Oberbayern



## Payment Process in the Mensa

Double Tap:

1. Eligibility verification of status group (Student, Employee, Guest)  
→ Discount will be applied
2. Pay with an Open-Loop Bank / Credit-Card





## Transition of credentials into the Smartphone Wallets

### Use Case: Physical Access Control (Access to learning spaces)

Physical access to the learning spaces (provided by the library)

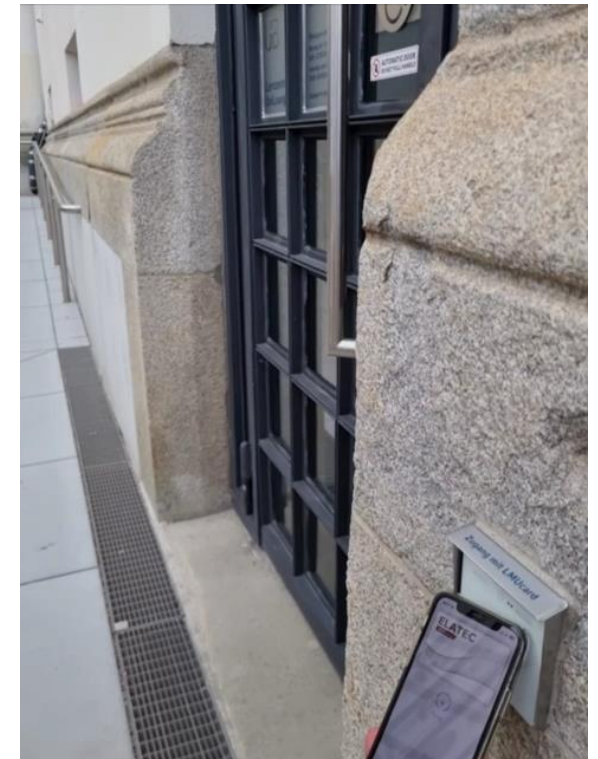


LMUcard

Access



Google Smart Tap



Apple



European University Alliance for Global Health

# eduTAP

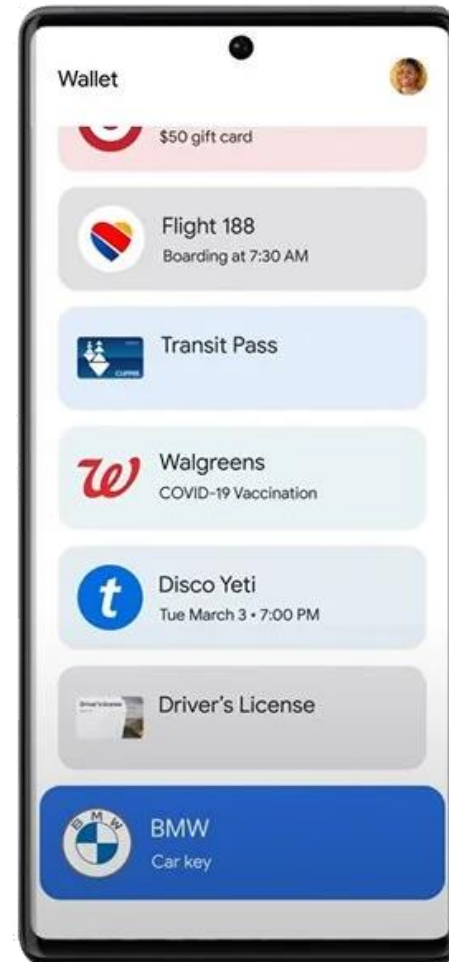
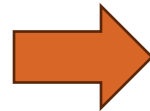
<https://eduTAP.eu>

*Modern Campus ID (Student ID / Staff ID)  
for universities in Europe & the world*

*Concept of eduTAP*



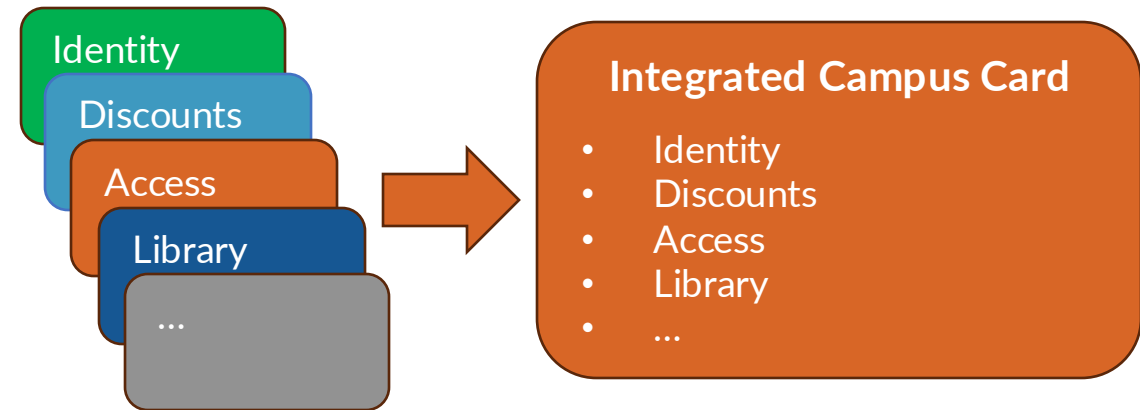
# Apple / Google Wallet – Focus on User Experience



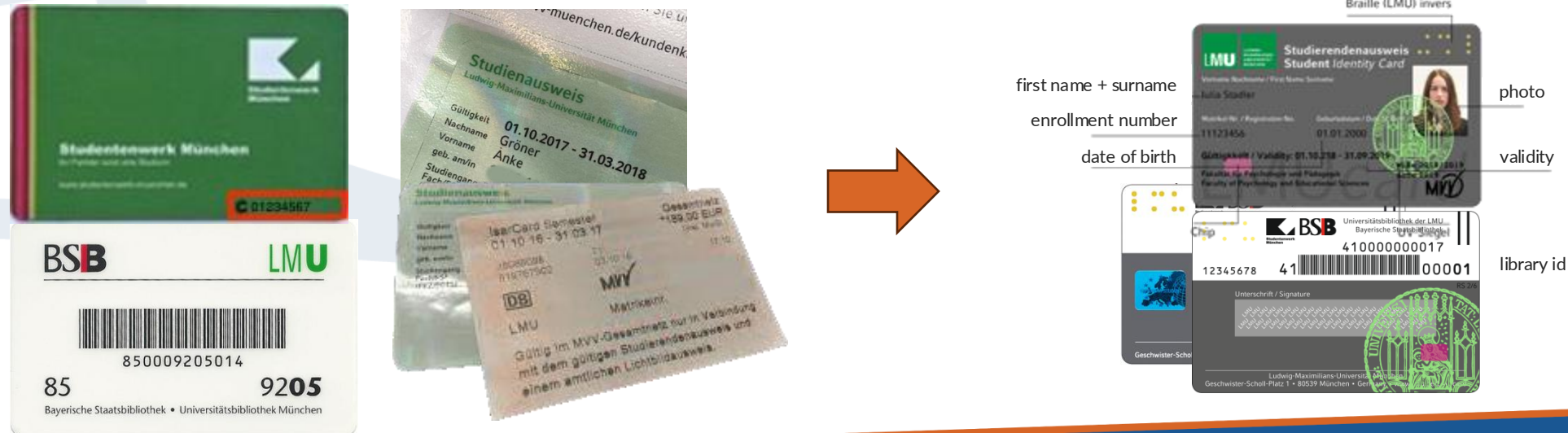
# Campus Cards from a User Experience Point of View

User preferences for **Physical Wallets**:

- ✓ Less cards are better
- ✓ **Contactless** cards are better



Example LMU 2015 – Transition from multiple cards into **ONE** integrated Campus Card

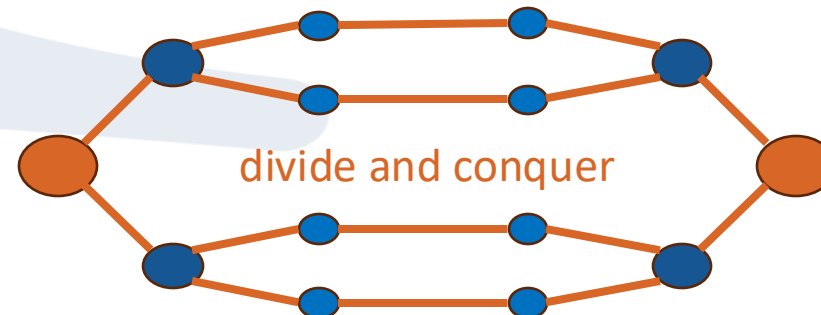
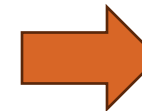
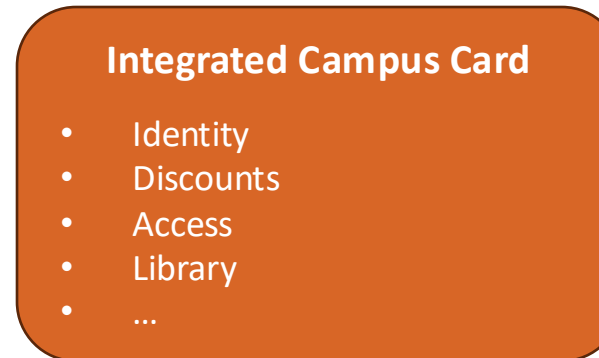




# Campus Cards from a User Experience Point of View

## User preferences for Smartphone Wallets:

- ✓ NFC / Contactless passes are better than scanning visual codes
- ✓ **One pass per service** is better
  - User control over which services to use
  - Auto-Presentation of pass needed
  - Clear contextual communication (additional information on pass specific for the service, specific branding of service)





# Digitizing cards **vs.** Leveraging the possibilities of smart devices → Stages of Digitalization – Period of Digital transformation

## Digitalization 1.0

- Digitization – create digital versions of analog documents

→ Digitization vs. Digitalization (machine readable content / payload)

## Digitalization 2.0

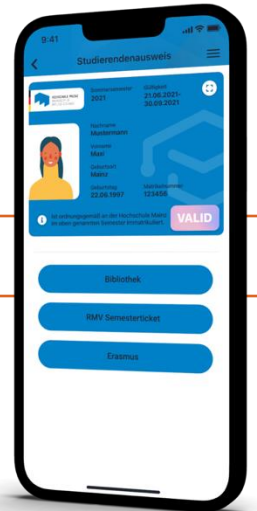
- Digitalization with internal focus
- Optimize internal administrative processes – **organization centric optimizations**
- Processing of data

→ Digital transformation (a hard shift)

## Digitalization 3.0

- Digitalization with focus on the user – **user centric optimizations**
- Optimization on how services are discovered and seamlessly used
- Flow of data

→ Service-oriented  
→ **Digital identity**  
→ Interoperability



→ User Experience

# eduTAP – educational tapping

## tap to ...

- ✓ identify yourself,
- ✓ verify your status,
- ✓ proof your entitlement,
- ✓ claim a discount,

## tap to access a service on-site directly:

- ✓ tap to pay (open and close loop-payments),
- ✓ tap to open a door,
- ✓ tap to lend a book,
- ✓ tap to take a campus shuttle.



*no app, just tap!*

<https://eduTAP.eu>

# eduTAP – educational tapping – What is eduTAP?

- eduTAP the concept
  - Splitting integrated campus cards into **dedicated service passes** and a **harmonized identity pass**
  - Leveraging the full potential of digital wallets
  - **Use existing**, secure and proven contactless **standards** – do not invent new standards

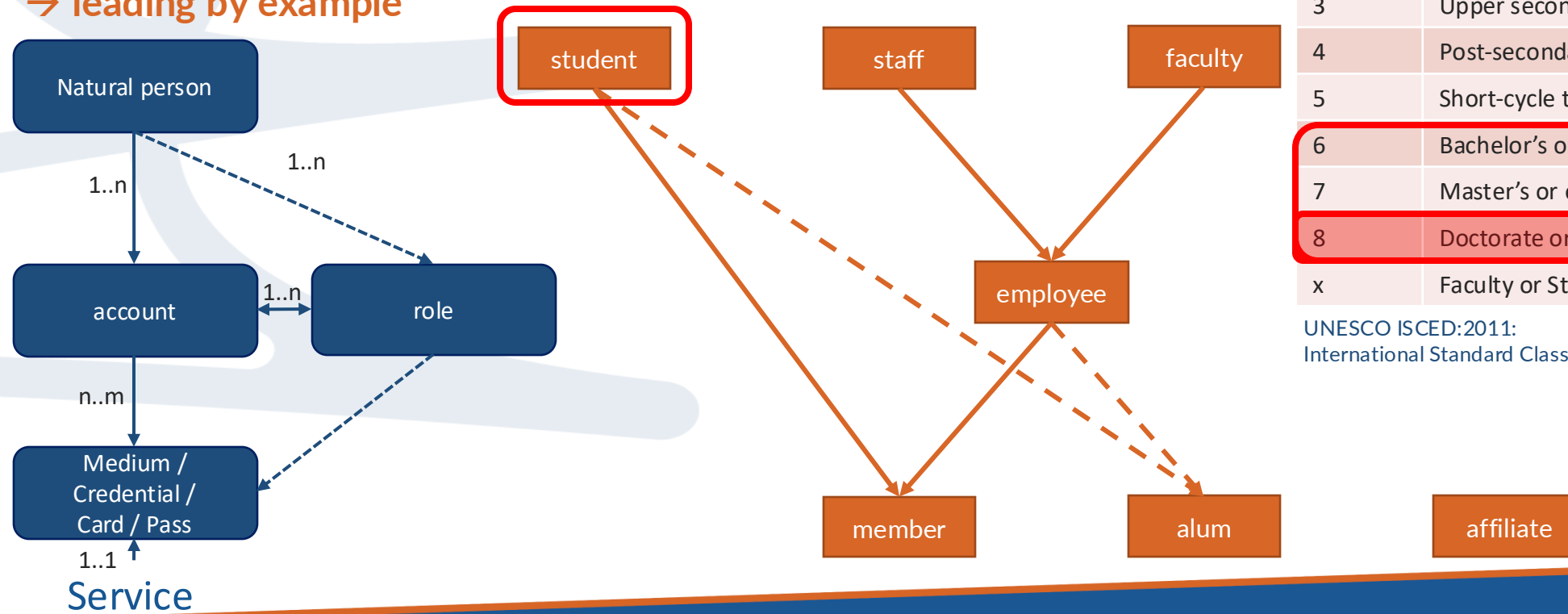
→ Focus on **User Experience** and **interoperability**

- eduTAP the software – to enable HEIs and Service Providers to:
  - create, issue, and manage passes within the smart device wallet
  - interact with wallet passes to give access to services, identify, and verify the status of students and staff members

→ eduTAP helps to fulfill the  
great vision of the European Student Card  
and **enables interoperability** and a **better user experience**

# Services and Service Consumers

- ✓ It is all about **providing services** → **eduTAP is for anyone**
  - ✓ For users of the (home or host) institution
  - ✓ User incoming for a short-term or long-term mobility
- ✓ Staff / faculty members are more likely on short-term mobility than students  
→ **leading by example**



## ISCED:2011 Level

| Level | Label                                 |
|-------|---------------------------------------|
| 0     | Early childhood education             |
| 1     | Primary education                     |
| 2     | Lower secondary education             |
| 3     | Upper secondary education             |
| 4     | Post-secondary non-tertiary education |
| 5     | Short-cycle tertiary education        |
| 6     | Bachelor's or equivalent              |
| 7     | Master's or equivalent                |
| 8     | Doctorate or equivalent               |
| x     | Faculty or Staff member of HEI        |

UNESCO ISCED:2011:  
International Standard Classification of Education

# eduTAP – educational tapping – for the users

eduTAP passes enable using smart devices to perform transactions with a

## simple tap

making everyday interactions **easier** and more secure both on campus and beyond.

## → User Experience

**Our mission** is to provide an **interoperable solution** that works for everyone

- for users of the (home or visiting) institution
- for user incoming for a short-term or long-term mobility
- staff / faculty members are more likely to be on short-term mobility than students

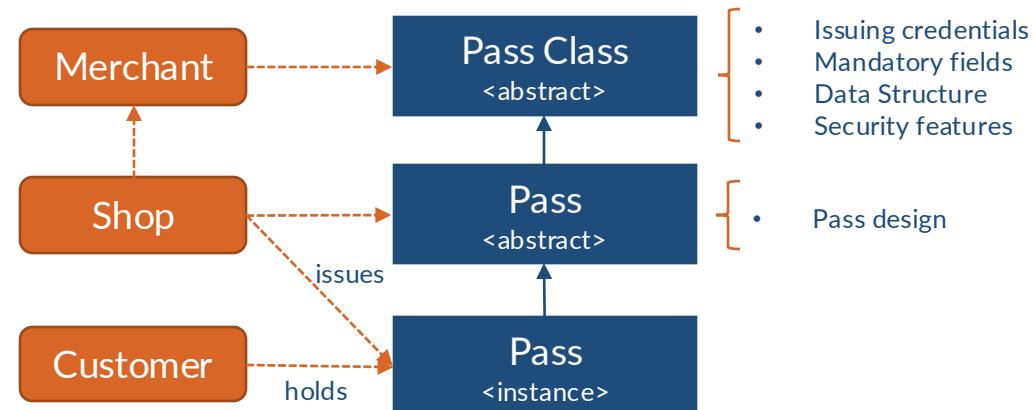


# Technical details and features of Wallets / Passes

The wallet is a **container for multiple passes**

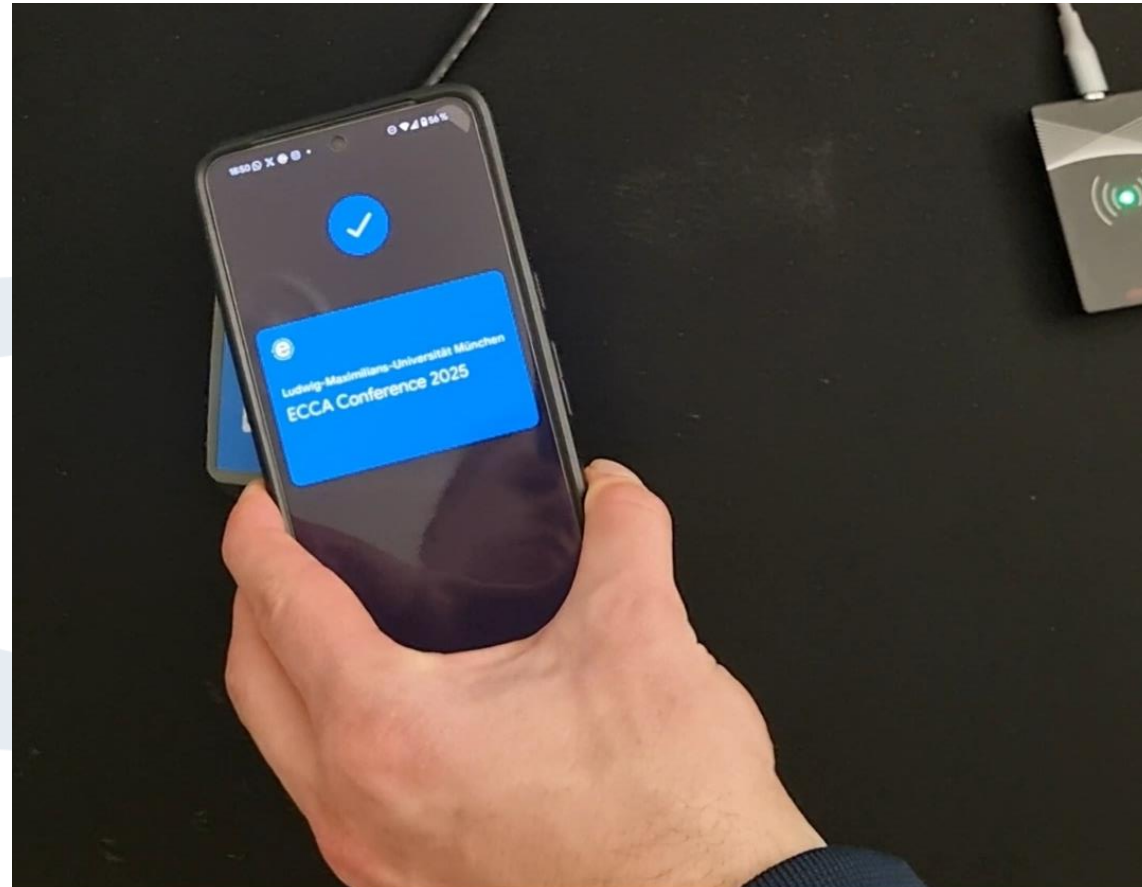
## Technical features:

- **Auto-Presentment / Smart Select** ← reader selects the specific pass to interact with (if available)
- Selectable Authentication requirements ← **security features**
- Express-mode (Apple)
- **Battery reserve** (pass is still usable if phone has powered off; important for keys, transit tickets, payment)
- Groupable
- Validity dates
- Remote updateable / revokable
- **Offline usable** (→ non traceable)
- *Host Card Emulation possible*



# Technical details and features of Wallets / Passes

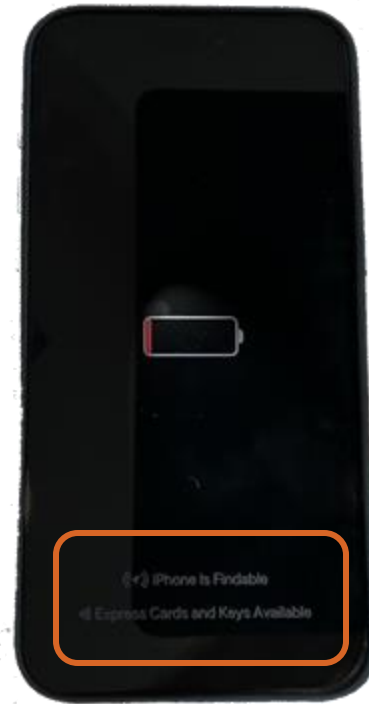
Auto-Presentment / Smart Select of an NFC enabled Pass in the Wallet



# Technical details and features of Wallets / Passes

## Battery Reserve Mode – Use the Wallet even if it's powered off

- for Keys (Dorm-Rooms, Class-Rooms, Building Access, Cars)
- for Payment
- for Transport Tickets

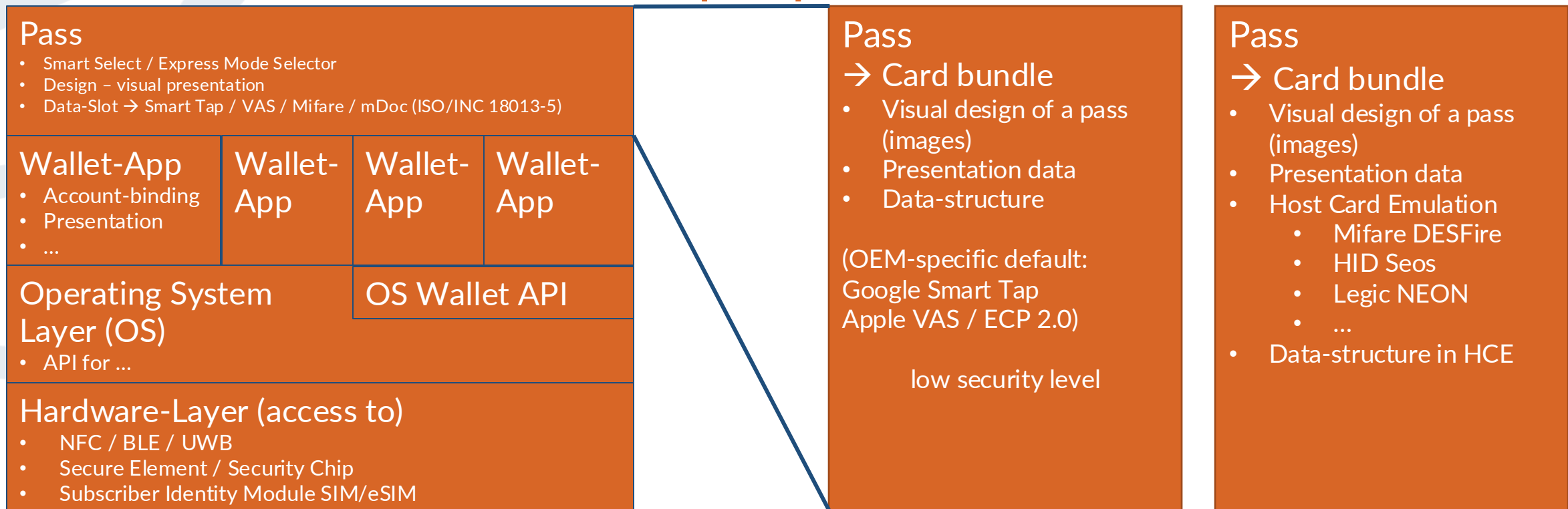


Up to 7 Days after shutdown of the smartphone

# Technical details and features of Wallets / Passes

- ✓ Technical details and features of wallets / passes → Path to an **interoperable** campus card
- ✓ If each pass (technology) can be individually selected and leveraged, and the pass in the wallet can be selected by the reader (**Smart Select / Auto-Presentment**)

→ **One pass per service**



# Technical details and features of Wallets / Passes

- For contact less communication is NFC the protocol
  - Communication via ISO 14443
  - Data Formats (File System access)
    - Google Smart Tap
    - Apple VAS (Value Added Services)
    - NFC ECP 2.0 (Enhanced Contactless Polling)
    - Host Card Emulation (HCE: of proprietary Protocols)
      - Mifare DESfire (via Mifare2GO / NXP secure element)
      - Legic NEON / Advant (via Legic Connect)
      - HID Seos
    - Internal the Secure Module acts as a JCOP smart card
  - ISO 18013-5 Personal Identification
    - NFC handover to BLE or WiFi
- Core functionality in Android and iOS OS API*



Security Level  
Chip / Smartcard

Security Level  
Smartphone

high

high

Legic NEON  
Legic Advant  
Mifare DESfire  
HID Seos

ISO 18013-5

ECP 2.0

Smart Tap  
VAS

low

Mifare Classic /  
Card UID

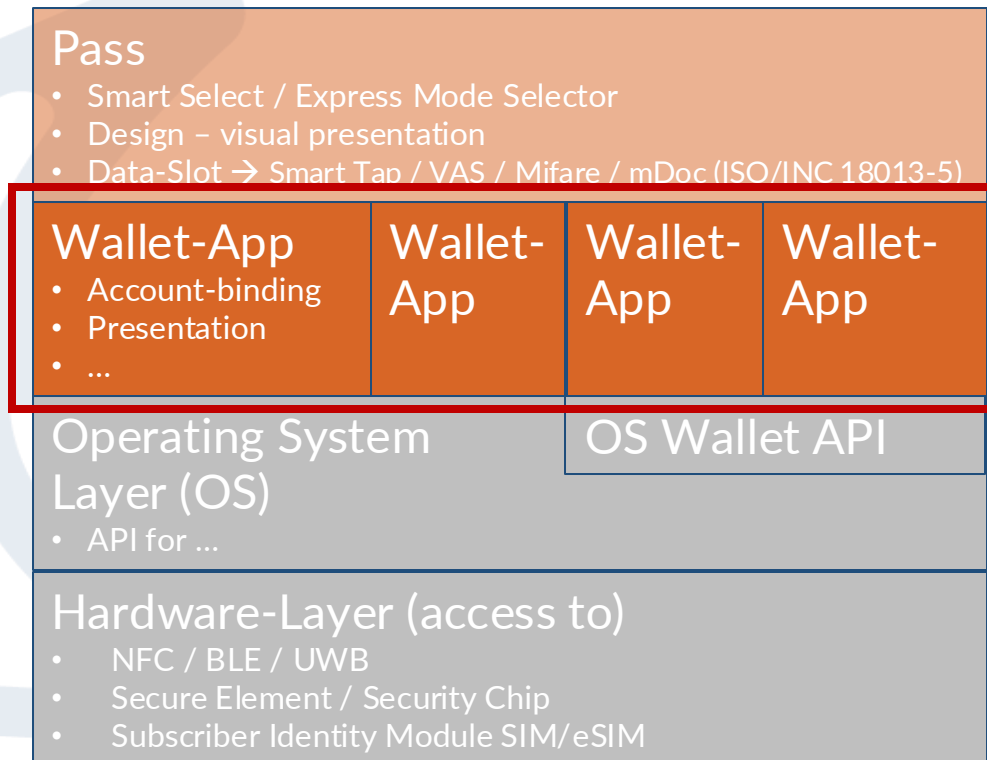
low

- Security consideration required by use case



# Technical details and features of Wallets / Passes

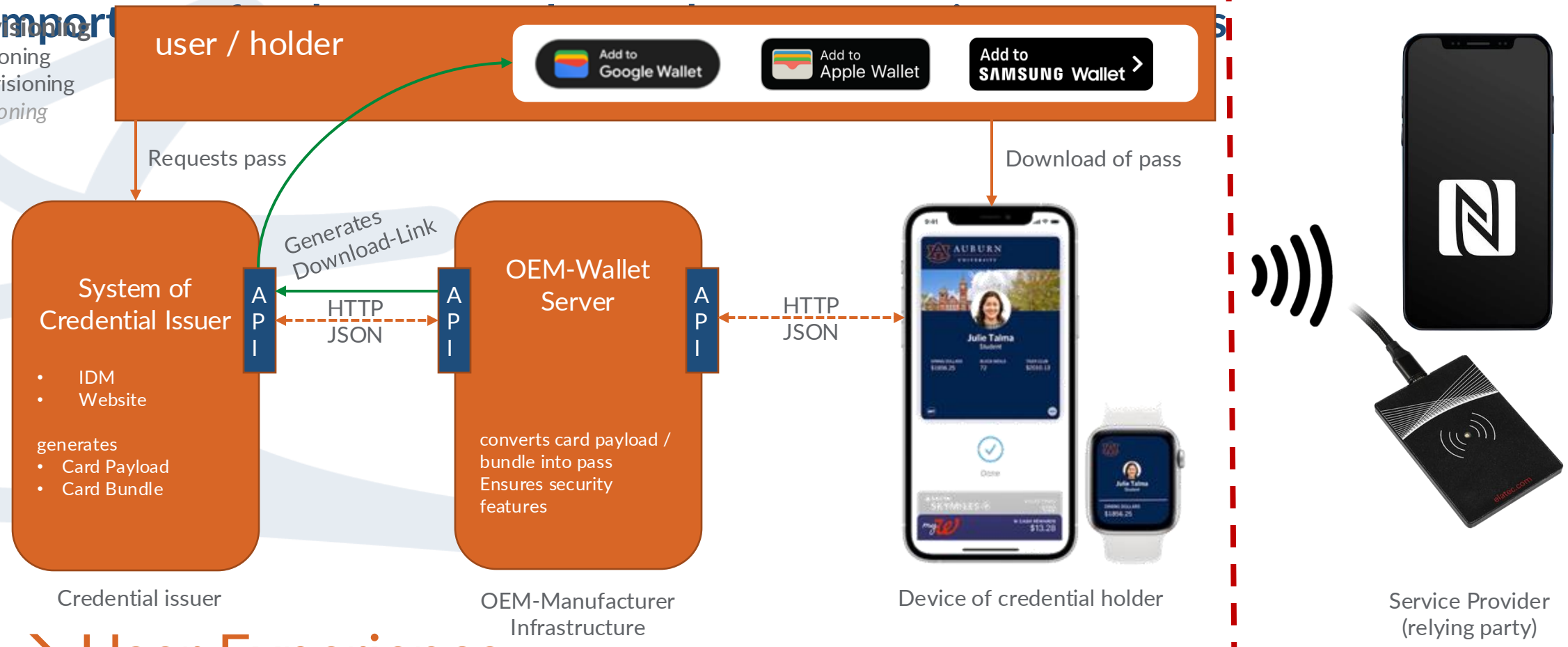
## Detail view on the Wallet



- The eduTAP solution is wallet **agnostic**, not bound to a specific wallet provider or company
- eduTAP just enables HEI and Service Providers to easily **create** and **issue** wallet passes

# Issuing process of wallet passes reedeming

- Web-Provisioning
- App-Provisioning
- Wallet-Provisioning
- Push-Provisioning



→ User Experience



European University Alliance for Global Health

# eduTAP

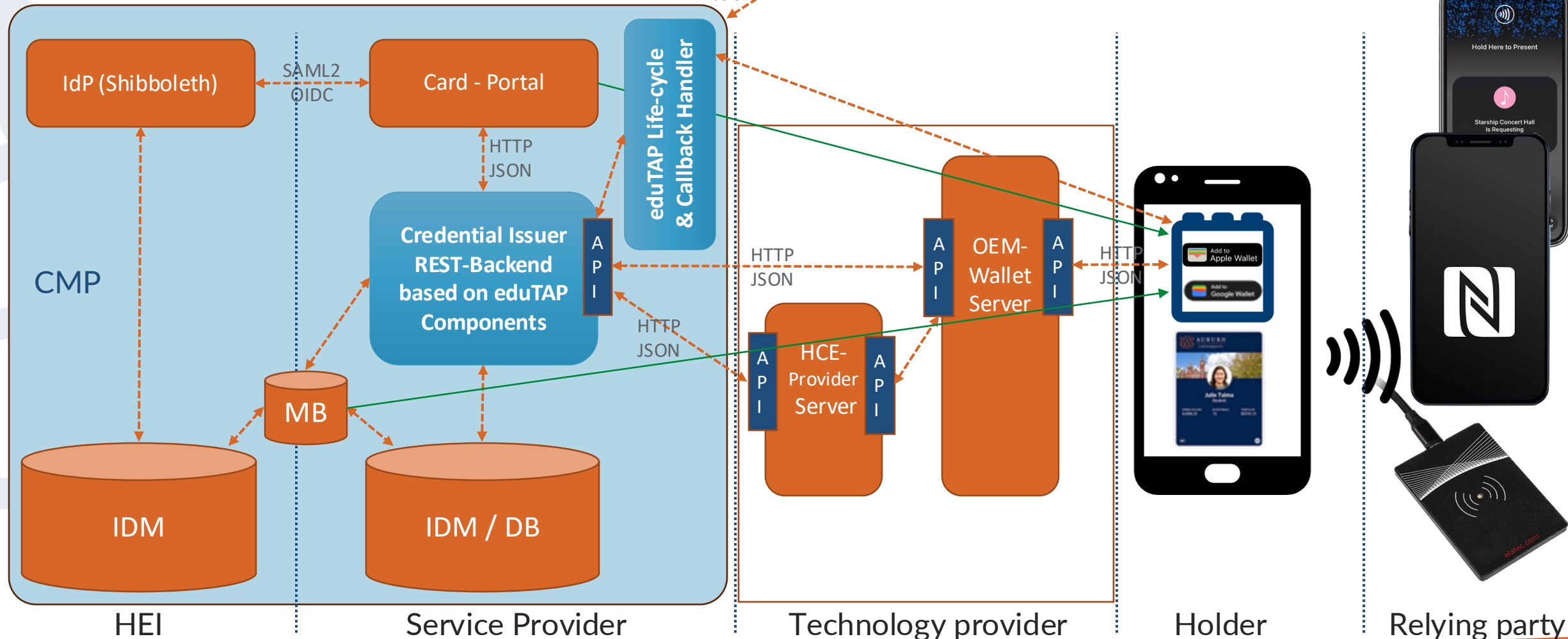
<https://eduTAP.eu>

*The Credential Management Platform in Detail*



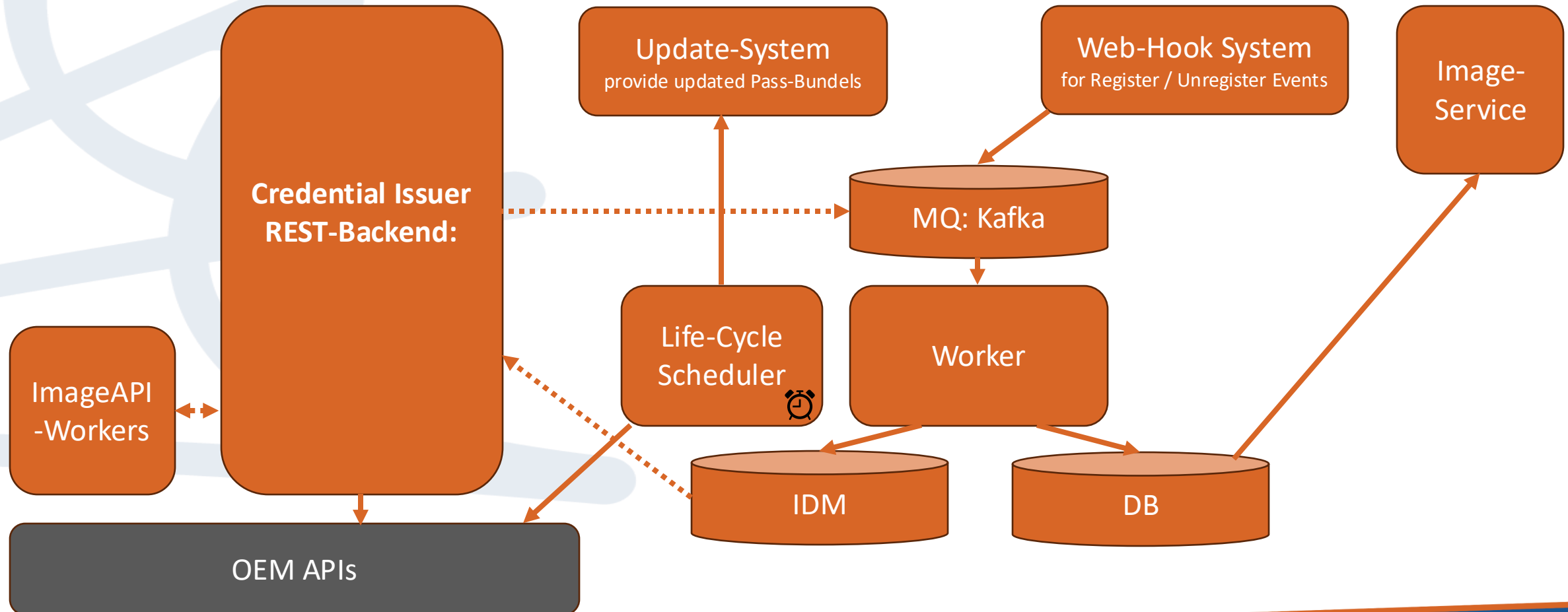
# Credential Management Platform and eduTAP Components

→ Architecture of the delivery system



# Credential Management Platform and eduTAP Components

→ Issuing, and Life-Cycle Management





# Credential Management Platform and eduTAP Components

→ eduTAP Components to the Frontend and to the Wallet APIs

**Credential Issuer  
REST-Backend:**

- To the Frontend
  - ✓ Authentication with SAML2 / OIDC / eIDAS
  - ✓ Identity-Linking with Apple, Google, Samsung, ...
- To the Wallets
  - ✓ edutap.wallet\_google
  - ✓ edutap.wallet\_apple
  - ✓ edutap.wallet\_Samsung
  - ✓ OID4VCI

# Credential Management Platform and eduTAP Components

Example of a simplified API with hardening and model validation: edutap.wallet\_google

## API Functions

Most tasks are done by import and using the sole `api` module:

```
from edutap.wallet_google import api
```

To tell the API functions what kind of item to deal with, the first parameter is the registered name of the model (except for `save_link`). Models can be the different top-level wallet-classes or -objects, but also issuers, permissions and such (see section models below).

|                                                               |                                                                            |
|---------------------------------------------------------------|----------------------------------------------------------------------------|
| <code>create</code> (data)                                    | Creates a Google Wallet items.                                             |
| <code>read</code> (name, resource_id)                         | Reads a Google Wallet Class or Object.                                     |
| <code>update</code> (data, *[, partial])                      | Updates a Google Wallet Class or Object.                                   |
| <code>message</code> (name, resource_id, message)             | Sends a message to a Google Wallet Class or Object.                        |
| <code>listing</code> (name, *[, resource_id, issuer_id, ...]) | Lists wallet related resources.                                            |
| <code>save_link</code> (models, *[, origins, iat, exp])       | Creates a link to save a Google Wallet Object to the wallet on the device. |

# Credential Management Platform and eduTAP Components

eduTAP Components are opinionated Software (*a Reference Implementation of the eduTAP Concept*)

- ✓  Pydantic for all Pass-Models
- ✓  HTTPX for API connections
- ✓  FastAPI for web-services
- ✓ Container based setup for deployment
- ✓ High SLA requirements, distributed, redundant, ...
- ✓ Reuses Concepts from the Zope / Plone (but with the modern Python possibilities)
  - Adapters and Multi-Adapters: Protocol and EntryPoints

```

1. @router_callback.post("/callback")
2. async def handle_callback(request: Request, callback_data: CallbackData):
3.     """FastAPI handler for the callback endpoint.
4.
5.     It is called by Google Wallet API when a user interacts with a pass.
6.     The callback is triggered on save and delete of a pass in the wallet.
7.     """
8.     # get the registered callback handlers
9.     try:
10.         handlers = get_callback_handlers()
11.     except NotImplementedError:
12.         raise HTTPException(
13.             status_code=500, detail="No callback handlers were registered."
14.         )
15.
16.     # extract and verify message (given verification is not disabled)
17.     callback_message = await verified_signed_message(callback_data)
18.     logger.debug(f"Got message {callback_message}")

```

```

1. @runtime_checkable
2. class CallbackHandler(Protocol):
3.
4.     async def handle(
5.         self,
6.         class_id: str,
7.         object_id: str,
8.         event_type: str,
9.         exp_time_millis: int,
10.         count: int,
11.         nonce: str,
12.     ) -> None:
13.         """
14.         :param class_id: ClassId
15.         :param object_id: ObjectId
16.         :param event_type: EventType
17.         :param exp_time_millis: Expiration time in milliseconds
18.         :param count: count retries
19.         :param nonce: nonce UID
20.
21.         :raises: ValueError
22.         """

```

→ Configurable Persistence Layers

# Credential Management Platform and eduTAP Components

## eduTAP vs Credential Management Platforms build on top of eduTAP

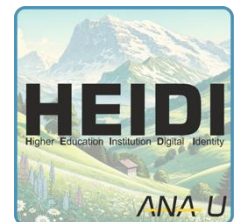
eduTAP → works for large Universities (Developers and Skills are available)

- ✓ eduTAP is Community Open Source (modelled after the Plone Community)
- ✓ eduTAP is a set of libraries and example implementations
- ✓ eduTAP does provide a set of templates for Pass Designs and necessary legal and technical documentation
- ✓ eduTAP does not provide a Frontend or Pass-Designer

Commercial Credential Management Platforms (CMP) → make or **buy**

e.g. HEIDI (Higher Education Institution Digital Identities) Platform is a CMP on-top of eduTAP-Components

- ✓ Provide Adapters to Business Applications (e.g. Campus-Management Systems, Library-Systems, ...)
- ✓ Provides a Pass-Designer
- ✓ Provides a Management-UI



# Credential Management Platform and eduTAP Components

## eduTAP Principles

- ✓ NO new standards, reuse existing proven standards
- ✓ Follow Identity & Access Management Principals of the eduGAIN AAI (Authentication and Authorization Infrastructure)
- ✓ Follow GDPR Principals – Data minimalization – Selective Disclosure – Zero Knowledge proves
- ✓ Security by Design
  
- ✓ All credential has a very short lifetime (max 7 days), but will be updated daily → eduPersonAssurance
- ✓ All credentials will have a registration and update web-hook configured
- ✓ Credentials will not be shareable by default, and are linked to a specific account or device



# Credential Management Platform and eduTAP Components

## Meet the eduTAP Team



**Alexander Loechel**

LMU Munich • EUGLOH •  
ECCA  
Project Lead • Developer



**Jose Filipe Alves**

U.Porto • EUGLOH • ECCA  
Expert on Credential Formats



**Morgan Persson**

Lunds universitet • EUGLOH •  
ECCA  
Expert on Card Reader and  
Credential Formats



**Simon Lund**

LMU Munich • EUGLOH  
Developer



**Christine  
Baumgartner**

HEIDI  
Project Management HEIDI



**Jens Klein**

HEIDI  
Developer



**Philip Auersperg**

HEIDI  
Developer



**Robert Niederreiter**

HEIDI  
Developer



European University Alliance for Global Health

# eduTAP

A *sneak peek* into the future of Student IDs

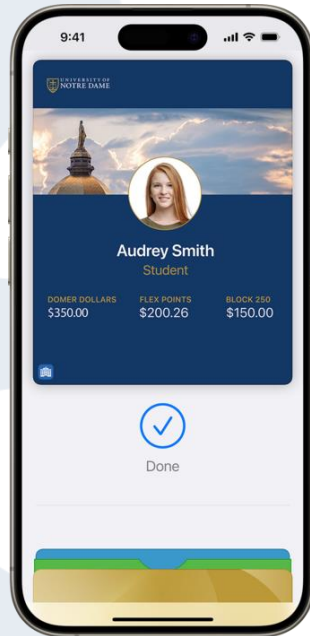
→ More interoperability and less passes by Verifiable Credentials

**eduTAP Common ID**



# eduTAP and the current Apple / Google Student IDs

How about the native, current **Student IDs** in Apple Wallet & Google Wallet (Mifare or HID based)?



- ✓ Fantastic integrated cards (**one card** for all services, perfect for **corporate**)
- ✓ *Should be used for all university services* (Apple's intention)
  - University / organisation-centric optimization
  - Does not support the “seamless mobility” requirement
  - Shared services, how about other users than own students?
  - **LESS secure** (if following the ESC data zone approach)
  - Not the required User Experience, and does not meet the legal requirements

This is a **dead-end solution**, it will not help or bring benefit, it actually provides a **WORSE User Experience**

It does NOT support **Selective Disclosure** and **Zero Knowledge Proof**

*Apple and Google is aware of this and working on it!*

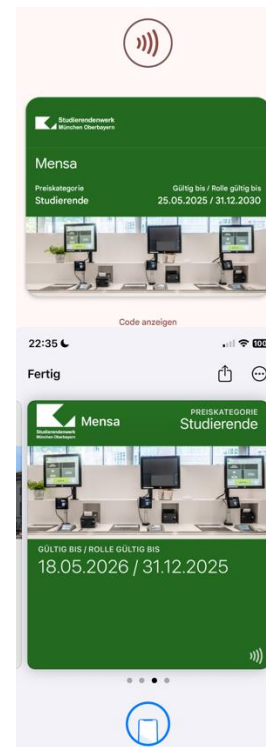
Apple and Google Access Platform are good for PAC and Payment passes

- **NOT** for Student or Staff IDs
- **GOOD** dedicated service passes

# Wallet Passes are about a great User Experience

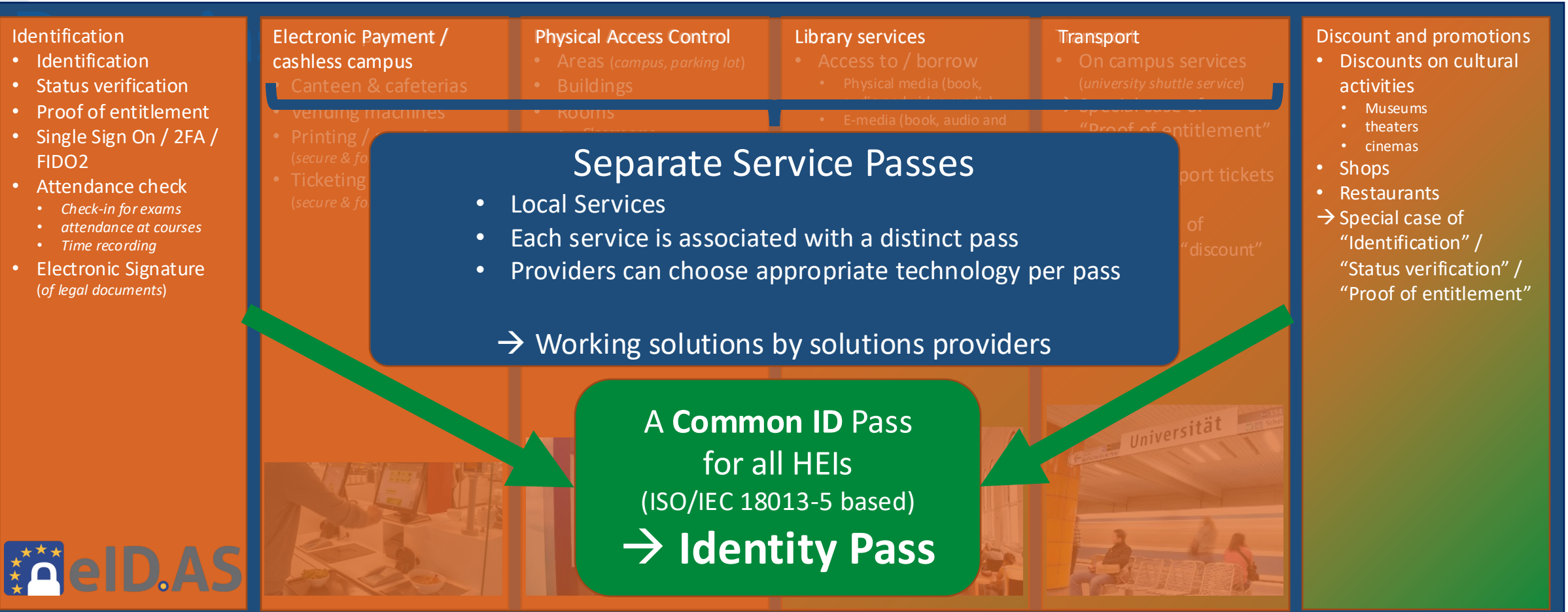
A **great User Experience** is key to adoption and usage

- ✓ Non-NFC Passes are **NO GOOD** User Experience
- ✓ Unnecessary passes should be avoided ← NO dedicated pass for simple eligibility verification (status, institution affiliation)



- ✓ Verifiable Credentials
- ✓ Selective Disclosure
- ✓ Zero Knowledge Proofs

# Think outside the card: Medium → Services



# Service Domains: Identity and Discount

From a legal point of view, Identity Documents of public universities are **official state photo IDs**

- ✓ Legal requirements
  - ✓ Security considerations
  - ✓ High assurance level
- **International standards** for HEI Identities → eduGAIN (incl. eduPerson, SCHAC)  
 e.g. European Student Identifier → routable information for course participation at other institutions (during mobility programmes)

## Identity Documents (Student ID, Staff ID and Affiliate IDs) are used

On-Site (proximity use-cases)

Online

On-Campus

Off-Campus

- ✓ Attendance at courses
- ✓ Check-in for exams
- ✓ Time recording
- ✓ Discounts in canteen & cafeterias
- ✓ Identification for university authorities (Student legal office and examination offices, campus security)

- ✓ Identification for public authorities (e.g., police)
- ✓ Identification for third party (e.g., checkin at hostels)
- ✓ Eligibility verification for discounts and access (e.g., role-based, age-based, institution related)

- ✓ As authentication factor



# A question of scale: Student IDs

## European Union

- ~ 5,000 Higher Education Institutions (HEIs)
- ~ 18,800,000+ tertiary students (2022)

## Erasmus+ Countries

- Includes EU + Iceland, Liechtenstein, Norway, North Macedonia, Serbia, Turkey
- Significantly expands the European higher education landscape → ~ 20,000,000+ tertiary students

## Worldwide

- ~ 18,000 recognized HEIs across 180+ countries
- ~ 200,000,000+ tertiary students globally

→ It should work everywhere: seamless mobility & easy service access  
→ a GLOBAL Student ID ←

→ Organized in  
National Research and Education Networks (NRENs)





# GÉANT – The IT provider for Higher Education in Europe

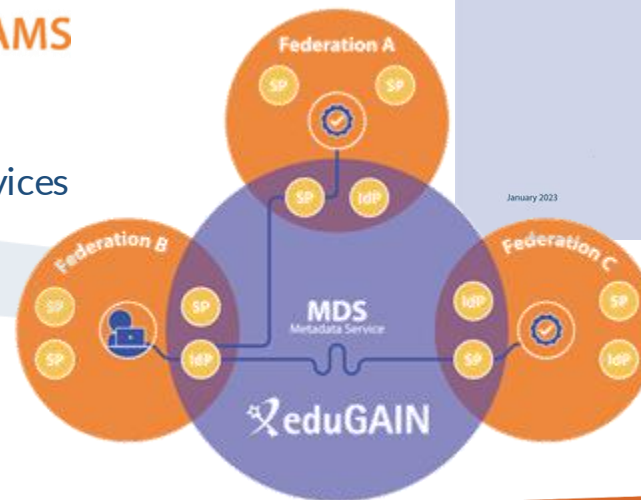
The GÉANT Association is the collaboration of European National Research and Education Networks (NRENs)

- Non-Profit Organization
- Service-Provider for IT / e-Infrastructure services for the European Higher Education Sector
- Pan European – Non EU members
- Key Services:

- **GÉANT Trust & Identity Services**



- GÉANT Network and Connectivity Services
- GÉANT Security Services
- GÉANT Cloud Services
- Procurement Agency for EU-HEIs



→ eduTAP is a perfect fit for GÉANT Trust & Identity Services

# Identity Pass: Reuse Shibboleth / eduGAIN data schema: The common authentication method in the educational sector

- ✓ Shibboleth – local or in federation (national or multi-national) → eduGAIN-AAI → SAML2 or OIDC



























































































































































LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN



DEUTSCHES FORSCHUNGSNETZWERK  
 AUTHENTIZIERUNGS- UND AUTORIZIERUNGS-INFRASTRUKTUR  
 LOGIN



HAW

## Informationsweitergabean MyAcademicID IAM Service

Beschreibung des Dienstes: *The MyAcademicID IAM Service provides Identity and Federated access management for the services of the European Student Card Initiative and the services directly supporting the digitisation of Erasmus+. The MyAcademicID IAM Service is provided by GEANT using the eduTEAMS service.*  
 Zusätzliche Informationen über diesen Dienst:  
<https://wiki.geant.org/display/SM/MyAcademicID+Identity+and+Access+Management+Service>

### An den Dienst zu übermittelnde Informationen

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anzeigename             | student                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Zugehörigkeit           | member<br>staff<br>employee                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Vertrauensgrad          | <a href="https://refeds.org/assurance">https://refeds.org/assurance</a><br><a href="https://refeds.org/assurance/IAP/low">https://refeds.org/assurance/IAP/low</a><br><a href="https://refeds.org/assurance/ATP/ePA-1m">https://refeds.org/assurance/ATP/ePA-1m</a><br><a href="https://refeds.org/assurance/ID/eppn-unique-no-reassign">https://refeds.org/assurance/ID/eppn-unique-no-reassign</a><br><a href="https://refeds.org/assurance/ID/unique">https://refeds.org/assurance/ID/unique</a><br><a href="https://refeds.org/assurance/ATP/ePA-1d">https://refeds.org/assurance/ATP/ePA-1d</a><br><a href="https://refeds.org/assurance/IAP/medium">https://refeds.org/assurance/IAP/medium</a><br><a href="https://refeds.org/assurance/profile/cappuccino">https://refeds.org/assurance/profile/cappuccino</a> |
| Berechtigung            | urn:maced:dir:entitlement:common-lib-terms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Personliche ID          | @lmu.de                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Zugehörigkeit           | staff@lmu.de<br>member@lmu.de<br>student@lmu.de<br>employee@lmu.de                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| eduPersonTargetedID     | HV6noX/05xHyzkH4z+vIGNdD80=                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| eduPersonUniqueID       | 7b719738cae7b120b6383786b47fe98@lmu.de                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Vorname                 | Simon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| E-Mail                  | @campus.lmu.de                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| o                       | uni-muenchen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| schacHomeOrganization   | lmu.de                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| schacPersonalUniqueCode | urn:cas:personalUniqueCode:int:sis:lmu.de:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Nachname                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Datenschutzinformationen dieses Dienstes

Die oben aufgeführten Informationen werden an den Dienst weitergegeben, falls Sie fortfahren.

**Wählen Sie die Dauer, für die Ihre Entscheidung zur Informationsweitergabe gültig sein soll:**

☐ Bei nächster Anmeldung erneut fragen.

☒ Erneut fragen, wenn sich die Informationen ändern, welche an den Dienste weitergegeben werden.

☐ Ich bin einverstanden, dass dieselben Informationen in Zukunft automatisch an diesen Dienst weitergegeben werden.

Diese Einstellung kann jederzeit mit der Checkbox auf der Anmeldeseite widerrufen werden.

Ablehnen

Akzeptieren

Impressum | Datenschutzerklärung | Kontakt und FAQ

[illegible]

# Verifiable Credential for HEIs (Prototypes)

European Student Card as Verifiable Credential



DC4EU non-foundational Ids

Table with available schemes for non-foundational IDs

| Scope                     | Data model name     | Brief explanation                                                                                          | Status/<br>Detailed explanation | Schema URL             | Registry URL                             |
|---------------------------|---------------------|------------------------------------------------------------------------------------------------------------|---------------------------------|------------------------|------------------------------------------|
| Non-foundational identity | EducationalID       | Identifies the natural person in the context of an educational organisation, including national extensions | Available                       | <a href="#">Schema</a> | <a href="#">Verifiable Data Registry</a> |
| Non-foundational identity | AllianceID          | Identifies a student or staff member as affiliated with a European university alliance                     | Available                       | <a href="#">Schema</a> | <a href="#">Verifiable Data Registry</a> |
| Non-foundational identity | EuropeanStudentCard | European Student Card for student mobility, based on DG-EAC's service                                      | Available                       | <a href="#">Schema</a> | <a href="#">Verifiable Data Registry</a> |
| Non-foundational identity | MyAcademicID        | Identity credential for student mobility, based on MyAcademicID and eduGAIN infrastructure                 | Available                       | <a href="#">Schema</a> | <a href="#">Verifiable Data Registry</a> |
| Non-foundational identity | ProfessionalID      | Identity credential for , based on                                                                         | Available                       | <a href="#">Schema</a> | <a href="#">Verifiable Data Registry</a> |
| Non-foundational identity | DoctorID            | Identity credential for , based on                                                                         | Available                       | <a href="#">Schema</a> | <a href="#">Verifiable Data Registry</a> |
| Non-foundational identity | EngineerID          | Identity credential for , based on                                                                         | Available                       | <a href="#">Schema</a> | <a href="#">Verifiable Data Registry</a> |

<https://github.com/dc4eu/educational-pilot/tree/main/sectorial-eaa-catalogue>



# Digital Identity in the Wallet

## Common Verifiable Credential Formats / Protocols for Digital Identities

- SD JWT (EU)
- ISO/IEC 18013-5 (mDL)
- W3C Verifiable Credential Data Model

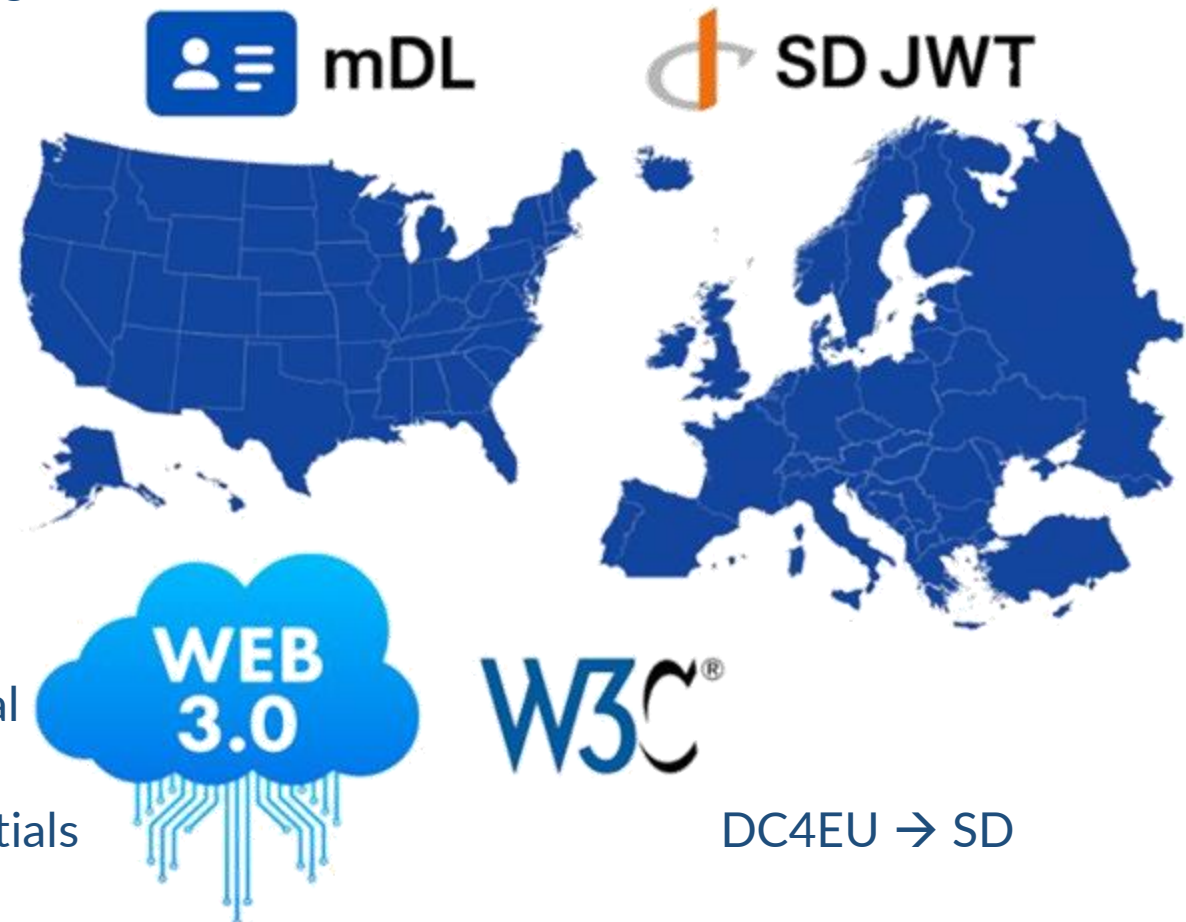
## Issuing and Redemption Protocols

- OID4VC (OID4VCI and OID4VP)
- ISO/IEC 18013-5

For **proximity** use-cases NFC / BLE transmission is crucial

→ Our choice ISO/IEC 18013-5 (mDL) for Wallet credentials  
JWT

Image Source: <https://www.linkedin.com/feed/update/urn:li:activity:7333379048112361472/>



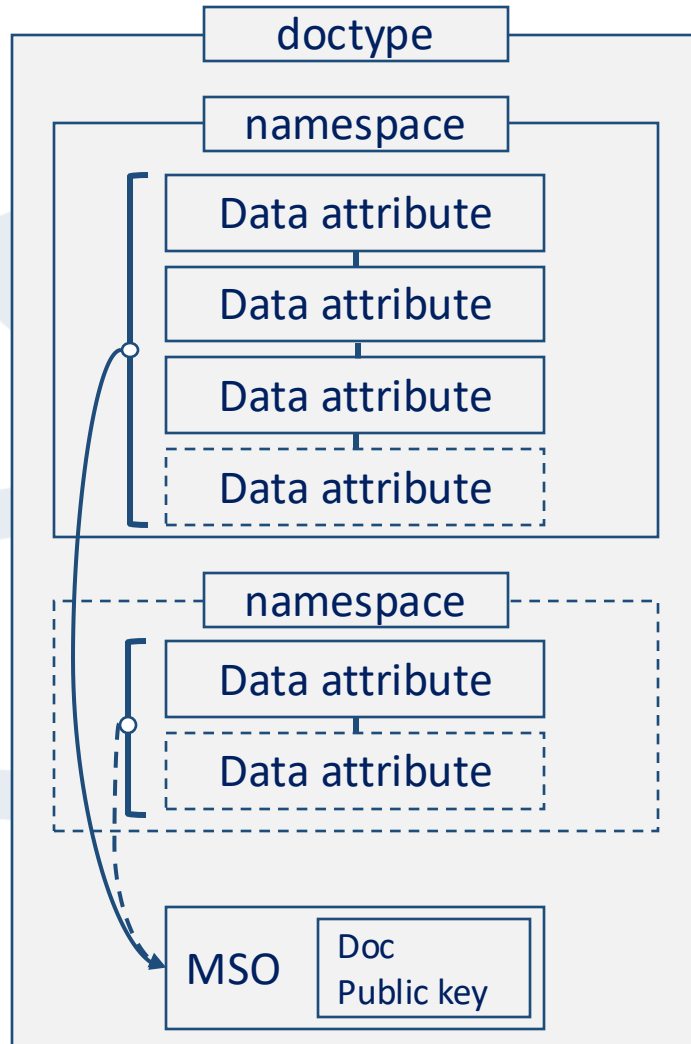
# A Common Identity Pass using the ISO/IEC 18013-5 model

- The “ISO/IEC 18013-5 Personal Identification” model is designed to support
  - A protocol for two devices to establish a **secure wireless communication** channel and exchange structured request and response message
  - **Identification** of the credential holder (**user binding**)
  - **Selective release of data elements** by the credential holder (**data minimization / consent**)
  - **Pre-consent** mechanism for frequently repeated processes (e.g., discount payment at Student Affairs Unions)
  - A protocol to retrieve credential data directly from the mobile device of the holder, **purely offline**, facilitating **availability** and **non-traceability**
  - An **optional** protocol to **retrieve additional data** from the issuing authority
  - A mechanism to establish **integrity** and **authenticity** of the credential data
  - A mechanism to confirm **device binding** (signing at transaction time)
  - Multi-document and **multi namespace aware** ← extendable for other use cases
- **Established governmental credential format that is already in use** ← **Following the ARF of EUDI-Wallet**  
→ mobile driving licence / eu.europa.ec.eudiw.pid.1 (prototype of national eIDs in Europa)
- **Already supported** by the OEM-Smartphone-Manufactures (Apple and Google)  
→ see: <https://android-developers.googleblog.com/2020/11/privacy-preserving-features-in-mobile.html>





# Identity Pass: ISO/IEC 18013-5 Data model



Reuse of directory data (common eduGAIN attributes) 

- portrait (photo, *mandatory for identification documents*)
- givenNames
- sn (surname)
- email
- eduPersonAffiliation / eduPersonScopedAffiliation
- eduPersonEntitlement
- eduPersonAssurance
- schacHomeOrganization
- schacPersonalUniqueCode → European Student Identifier (ESI) - MyAcademicID
- schacDateOfBirth
- schacYearOfBirth
- schacPlaceOfBirth
- schacExpiryDate
- ...

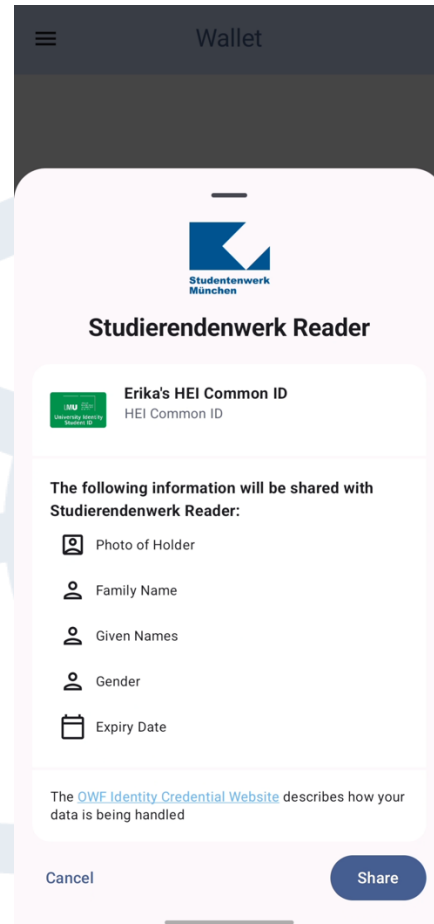


*works like a Shibboleth authentication  
on-site → trained processes*

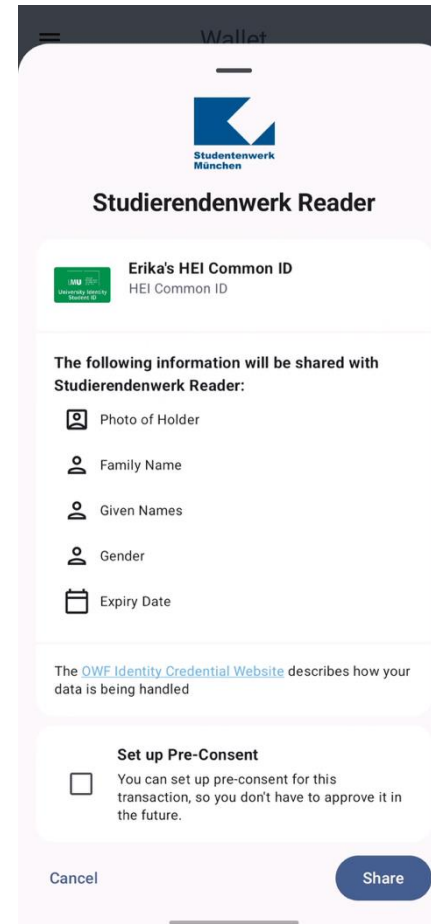
see also DC4EU education ID

(<https://code.europa.eu/ebsi/json-schema/-/tree/main/schemas/vcdm1.1/education/verifiable-education-id>)

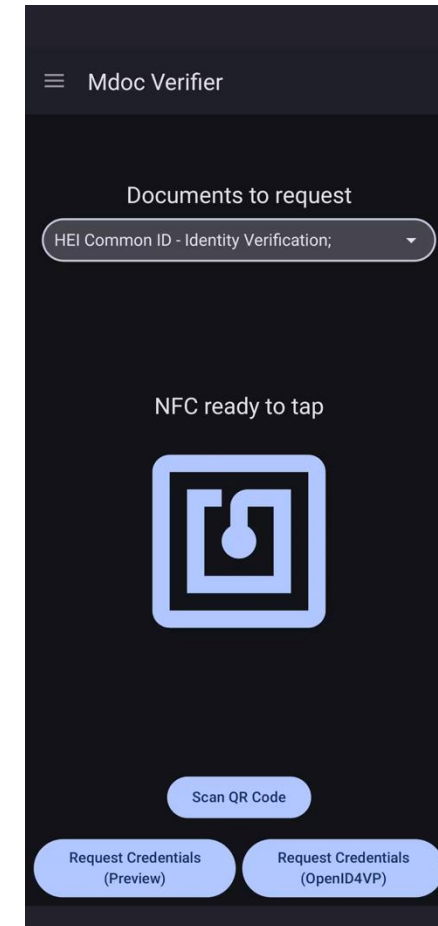
# Identity Pass: Path to the best User Experience and Interoperability



Informed Consent



Informed Consent  
+ Pre-Consent



Verifier App

# Campus Cards – Point of Views

Technology Experts vs. Consumers (Students, staff and professionals)

The Technology and issuing processes are not important to the consumers (card / pass holders)

The IDs and service passes should:

- ✓ **just work – service access must be commodity**
- ✓ be immediately available when needed
- ✓ be secure & GDPR compliant
- ✓ trustworthy
- ✓ **accepted everywhere**
- ✓ *be boring – no need for attention – user experience matters*

Students, teachers and researches should be able to focus on the important things,

→ their studies and research

→ enjoy student mobility (and do not need to wait weeks for a physical campus card)

# “Boring” is a good thing

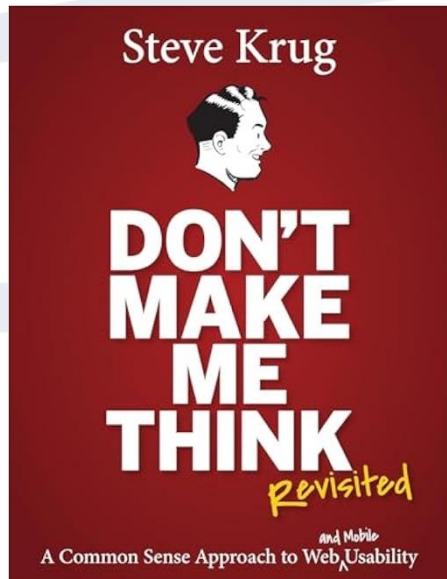
Every company gets about three innovation tokens.

Source: Dan McKinley, "Choose Boring Technology"

<http://mcfunley.com/choose-boring-technology>

“Boring” let you get things done

Same is true for people, if you **need to think** to get something done it costs effort, and energy for other more important things



Don't make me Think

→ User Experience is important

You should **NOT** need to care about “boring” and “un-important” stuff

You should be able to **focus on the important** things  
(studying, learning, researching, ...)





European University Alliance for Global Health

# eduTAP

<https://eduTAP.eu>

*Modern Campus ID (Student ID / Staff ID)  
for universities in Europe & the world*





# Demo → Download an eduTAP created pass



<https://demo.edutap.eu/>



Add to Google Wallet



Add to Apple Wallet

# eduTAP – educational tapping

## tap to ...

- ✓ identify yourself,
- ✓ verify your status,
- ✓ proof your entitlement,
- ✓ claim a discount,

## tap to access a service on-site directly:

- ✓ tap to pay (open and close loop-payments),
- ✓ tap to open a door,
- ✓ tap to lend a book,
- ✓ tap to take a campus shuttle.

TAP TO PAY



*no app, just tap!*

<https://eduTAP.eu>

# eduTAP – educational tapping – What is eduTAP?

- eduTAP the concept
  - Splitting integrated campus cards into **dedicated service passes** and a **harmonized identity pass**
  - Leveraging the full potential of digital wallets
  - **Use existing**, secure and proven contactless **standards** – do not invent new standards

→ Focus on **User Experience** and **interoperability**

- eduTAP the software – to enable HEIs and Service Providers to:
  - create, issue, and manage passes within the smart device wallet
  - interact with wallet passes to give access to services, identify, and verify the status of students and staff members

→ eduTAP helps to fulfill the  
great vision of the European Student Card  
and **enables interoperability** and a **better user experience**

## We'd love to hear from you!

### Get in touch:

eduTAP Project (<https://eduTAP.eu/>)

- Alexander Loechel (LMU Munich)
- Simon Lund (LMU Munich)
- José Filipe Alves (U.Porto)
- Morgan Persson (Lunds universitet)

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